Bank Market Power and Interest Rate Risk *

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November 13, 2026

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Abstract

How does relative market power in funding and lending shape bank interest rate risk exposure and monetary transmission? We document a novel pricing fact: banks with less rate-sensitive funding charge higher fixed-rate loan spread over variable-rate loans, despite stable funding makes fixed-rate lending an attractive earnings hedge. Greater loan market concentration is associated with higher spreads conditional on funding sensitivity and a stronger expansion of fixed-rate credit after monetary tightening. We develop a two-date model in which banks trade higher expected earnings from restricting fixed-rate lending against greater earnings risk due to forgone hedging. A restriction on relative market power reproduces the negative premium–sensitivity slope, while endogenous entry explains why competition in both markets moves together. A calibrated investment extension captures lower premiums and a higher fixed-rate share after an interest rate increase. The extension implies a contraction in aggregate investment. An application shows that greater bank risk-bearing capacity can raise expected earnings and earnings volatility while substantially reducing fixed-rate lending but barely changing aggregate investment. Banks that can bear more risk become less willing to supply borrower protection, linking bank risk management to monetary transmission.

*I am grateful to Wei Cui for his guidance and support throughout this project. I also thank Adrien Couturier, Thomas Lazarowicz, Marta Morazzoni, Alan Olivi, Mert Onen, Manuel Paillacar-Almonacid, Franck Portier, Lukasz Rachel, Morten Ravn, Guido Spano, Vincent Sterk, Pengfei Wang, Yue Wang, Nianyi Zhang, and seminar participants at UCL for their helpful comments and discussions.

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1 Introduction

Monetary policy affects output and investment through the cost and availability of bank credit. Yet, its transmission also depends on how banks manage interest-rate risk and allocate it between themselves and borrowers. By adjusting the pricing and composition of fixed- and variable-rate lending, banks influence borrowers' exposure to interest-rate changes, their investment decisions, and ultimately aggregate economic activity. These decisions depend on banks' market power on both sides of their balance sheets. Deposit market power determines how funding costs respond to interest rates, while lending market power shapes banks' incentives to supply fixed-rate protection. How does the interaction between funding and lending market power shape banks' interest-rate risk management and monetary transmission?

We link U.S. regulatory data on bank funding and detailed loan-level data, and document a novel pricing fact: banks with less rate-sensitive funding charge a higher fixed-rate premium, the difference between the origination rates on fixed- and variable-rate loans. A one-standard-deviation decrease in interest expense sensitivity is associated with a 5.3-basis-point higher fixed-rate premium. This pattern is puzzling: banks with stable funding costs should value fixed-rate lending as an earnings hedge and therefore charge *lower* premiums. Yet the data show precisely the opposite.

Empirical evidence points to the joint roles of funding and lending market power in this pricing relationship. We show that banks with more rate-insensitive funding operate in more concentrated local lending markets. We find that a one-standard-deviation increase in bank lending market concentration is associated with a 22.5-basis-point higher premium, holding funding sensitivity fixed. Empirical results are also consistent with bank hedging incentive: higher funding rate-sensitivity is associated with a higher premium, conditional on lending market concentration. These findings suggest that the fixed-rate premium reflects both funding and lending market power.

Relative market power also impacts loan quantities and composition. Following monetary tightening, fixed-rate loan stocks expand while variable-rate stocks contract. The expansion in fixed-rate lending is stronger at banks with greater lending market power, after allowing shock responses to vary with interest expense sensitivity.

To interpret these facts, we build a model with interest-rate uncertainty. The model features risk-averse banks choosing the composition of lending and entrepreneurs selecting fixed- or variable-rate loans. Depositors search and compare offers locally, generating funding-rate passthrough that increases with deposit competition. Banks compete in quantities in a wider fixed-rate lending market, while variable-rate lending is competitive.

The central mechanism is that *relative* market power in funding and lending determines banks' chosen exposure to interest rate risk. Funding market power governs interest expense sensitivity and hence the fixed-rate loan share that hedges earnings. Lending market power

can make it profitable to restrict fixed-rate lending and sustain a larger spread over variable-rate loans. Banks thus trade higher expected earnings against forgone hedging and potential losses following adverse interest rate movements. The key result is that, when lending market power is sufficiently strong relative to funding market power, the model generates a negative premium–sensitivity slope, in line with our stylised fact. Importantly, the model reveals how lending market power impairs risk-sharing between banks and borrowers, with a welfare cost that is *non-monotonic* in interest rate volatility.

We further endogenise bank entry to explain the joint variation in funding sensitivity and lending competition. Larger markets attract more entrants by allowing banks to spread fixed participation costs over greater business volume. Entry increases both deposit-rate sensitivity and competition in fixed-rate lending, with opposing effects on premiums: the former weakens the hedging incentive, while the latter reduces rents from restricting fixed-rate credit. Therefore, in the symmetric entry benchmark, market size can select the degree of competition along the premium–sensitivity relationship derived above.

We embed this mechanism in an equilibrium model of investment and household saving, where fixed-rate protection from interest rate risk supports larger entrepreneurial projects. A higher benchmark rate initially widens deposit margins and attracts bank entry. Greater lending competition makes fixed-rate protection cheaper; at the same time, the entry brings more responsive funding costs, weakening banks' incentive to provide fixed-rate loans as an earnings hedge. Allowing banks to hedge externally at a constant marginal cost, and calibrating two markets to observed financial conditions and their responses to monetary tightening, we find that the competition effect dominates the hedging effect in the aggregate: borrowers shift towards fixed-rate finance even as higher borrowing costs reduce project sizes. Following a 50-basis-point rate increase, fixed lending rises by 6.27%, while total investment and household deposits fall by 0.67%. Thus, an expansion of fixed-rate credit can coexist with a contraction in total loans and investment, consistent with data. The price of protection governs the composition of borrowing and cushions the investment response, but does not overturn the effect of higher financing costs.

Finally, the same calibration shows that greater bank risk-bearing capacity primarily changes the composition of credit, with little effect on aggregate lending or investment. Banks that can tolerate more earnings risk have less reason to supply fixed-rate loans as a hedge and use their lending power to charge more for protection. Holding bank counts and expected rates fixed, an 80% increase in effective capacity raises the fixed-rate premium by 7.58 basis points and lowers the aggregate fixed share by 7.13 percentage points. Fixed lending falls by 38.38%, but floating lending rises by 8.74%, leaving total lending and investment only 0.02% lower. Borrowers mainly switch contracts, while wage adjustment moderates the change in project scale.

This result identifies a limit to policies that seek to improve borrower risk sharing by

increasing banks' capacity to bear risk. Diversification or anticipated public support may raise effective capacity, but banks need not use it to insure borrowers: weaker hedging incentives can instead make fixed-rate protection more expensive and shift interest rate risk towards entrepreneurs. Through this channel, greater capacity delivers little aggregate credit stimulus and can reduce borrower protection. Its effectiveness therefore depends on banks' incentives to supply insurance, shaped by relative funding and lending market power, as well as their ability to absorb risk.

Relationship to literature. Our paper contributes to three strands of literature: theories of bank interest rate risk, bank market power, and risk sharing through loan contracts.

Theories of bank interest rate risk. Hellwig (1994) studies the efficient allocation of interest rate risk under liquidity provision, showing that eliminating exposure can sacrifice reinvestment opportunities. Drechsler et al. (2021) show that fixed-rate assets can stabilise earnings when deposit franchises insulate funding costs from market rates. Di Tella and Kurlat (2021) explain exposure through a dynamic hedging motive: higher rates raise deposit-intermediation profits, making banks willing to accept contemporaneous capital losses. Varraso (2025) links endogenous maturity choices to bank net worth and financial constraints in a competitive intermediation model.

We identify a complementary mechanism through loan-market pricing power. Banks trade the earnings hedge from fixed-rate lending against the rents preserved by restricting its supply. Relative funding and lending market power governs this tradeoff, linking banks' chosen exposure to the price of borrower protection. Our focus on earnings exposure differs from the market-value sensitivity of bank franchises emphasised by DeMarzo et al. (2024).

Bank market power. A large and growing literature studies bank market power on both sides of the balance sheet. Funding-market power shapes monetary transmission (Drechsler et al., 2017) and financial fragility (Egan et al., 2017), while lending-market power affects mortgage-rate pass-through and refinancing (Scharfstein & Sunderam, 2016), as well as corporate borrowing rates and aggregate activity (Bocola et al., 2026). Wang et al. (2022) jointly model deposit and loan competition to study monetary transmission. Hachem and Song (2021) show how interbank market power shapes liquidity choices and can make tighter liquidity regulation generate a credit boom.

We contribute by examining how funding and lending market power jointly shape banks' interest rate risk management, motivated by our evidence that lending concentration is associated with higher fixed-rate premia conditional on funding sensitivity. We combine deposit market power arising from search frictions, as in Choi and Rocheteau (2023) and Head et al. (2025), with Cournot competition in lending. Endogenous bank entry then provides a common source of variation in competition across the two markets. Related evidence on the importance of entry comes from Begenau and Stafford (2025), who show

that branch expansion shapes deposit growth during monetary tightening.

Loan contracts and risk sharing. Fixed- and variable-rate loans allocate interest rate risk between banks and borrowers (Santomero, 1983). Vickery (2008) finds that financially constrained small firms are more likely to borrow at fixed rates, consistent with insurance demand. Kirti (2020) shows that banks with more rate-sensitive liabilities supply more floating-rate credit and price it more favourably relative to fixed-rate credit. Our evidence points to the opposite pricing relationship. Differences in lending competition may reconcile these findings: Kirti's pricing evidence uses residential mortgage quotes, whereas we study small business loans. Our model reproduces both relationships by varying relative funding and lending market power, highlighting its role in allocating interest rate risk.

Liu and Elenev (2025) connect mortgage fixation to financial stability and risk sharing; Abad et al. (2026) examine how loan fixation and bank solvency shape monetary transmission; and Bretscher et al. (2018) link interest rate uncertainty to corporate hedging and investment.

We extend these accounts of borrower insurance demand and banks' funding incentives by jointly determining fixed-rate loan pricing, borrower contract choice, and investment in our quantitative model. Greater bank risk-bearing capacity weakens the incentive to hedge earnings through fixed-rate lending. Banks restrict fixed credit and raise premiums, accepting greater earnings risk for higher expected earnings. More expensive protection reduces investment. This mechanism connects banks' capacity to bear risk to the supply of borrower protection and real activity.

2 Stylised Facts

Our empirical analysis combines loan-level records from the U.S. Small Business Administration's (SBA) 7(a) programme, bank regulatory reports (Call Report), and the Federal Deposit Insurance Corporation's (FDIC) Summary of Deposits. We consolidate affiliated institutions at the ultimate holding-company level, allowing for funding and asset allocation to be coordinated across subsidiaries through internal capital markets (Campello, 2002). Throughout the paper, "bank" refers to this consolidated entity.

The SBA records provide detailed loan-level information, including each loan's rate type and interest rate at origination. They offer a detailed window into small-business lending, a quantitatively important segment of bank credit: loans below $1 million accounted for 22% of domestic banks' commercial and industrial loan origination.[1] Small firms' dependence on bank finance and limited use of derivatives make their loan terms particularly consequential

[1]By value in May 2017. Calculated from Table 2 of the Federal Reserve's *May 1–5, 2017 Survey of Terms of Business Lending*. The relevant category covers loans of $10,000 to below $1 million; loan size serves as a proxy for small-business lending.

for managing interest rate exposure (Vickery, 2008). These features make small-business lending a useful setting for studying lending market power.

Call Report provides bank-level funding information, including interest expense, whose sensitivity to interest rates we use to assess funding market power following Drechsler et al. (2021). Call Reports also provide information on the repricing profile of outstanding loans. For the quantity analysis, we follow Kirti (2020) and proxy variable-rate loan stocks by loans maturing or repricing within one year, and fixed-rate stocks by the remaining loans. From the Summary of Deposits, we construct a county-level branch-count Herfindahl–Hirschman index (HHI) by summing banks' squared shares of local branches. We average these county indices equally across the counties in each bank's branch network to measure the local banking concentration it faces. Given the local nature of SBA lending,[2] we use this concentration measure as a proxy for lending market power, once conditional on funding sensitivity.[3] Appendix A provides detailed descriptions of the data and variable construction.

We document two stylised facts about the pricing and quantity of fixed-rate credit. First, banks with less rate-sensitive funding charge higher fixed-rate premiums. In the loan-level regressions, greater local banking concentration is associated with a larger fixed–rate premium, conditional on funding sensitivity and observed loan and bank characteristics. Second, fixed-rate loan stocks expand following monetary tightening, while variable-rate loan stocks initially contract. The expansion of fixed-rate lending is larger at banks operating in more concentrated markets. Together, these patterns point to the relative funding and lending market power, as a determinant of the pricing and provision of fixed-rate credit.

2.1 Fixed-rate premium and relative market power

Banks with less rate-sensitive funding charge higher fixed-rate premia. Figure 1(a) shows that a one-standard-deviation increase in interest expense sensitivity, β_b^{exp}, is associated with a 5.3-basis-point lower premium. This relationship runs against the prediction from earnings hedging alone. When funding costs respond little to interest rates, fixed-rate lending helps stabilise bank earnings, giving banks an incentive to offer these loans at lower premia. The observed negative slope therefore suggests that other pricing incentives vary systematically with funding sensitivity.

One possible explanation lies in differences in lending competition. Figure 2(a) maps county-level concentration measured using banks' shares of SBA 7(a) loan dollars and reveals substantial geographic variation. Some local markets have lending distributed across many banks, while others concentrate lending among relatively few providers. This variation points

[2]Di and Pattison (2020) report that 71% of loans in their matched 2017 SBA sample went to borrowers located within ten miles of a branch of their lender.

[3]Funding sensitivity therefore reflects broader funding conditions, while branch-network concentration captures local competition for borrowers. Appendix A develops this interpretation.

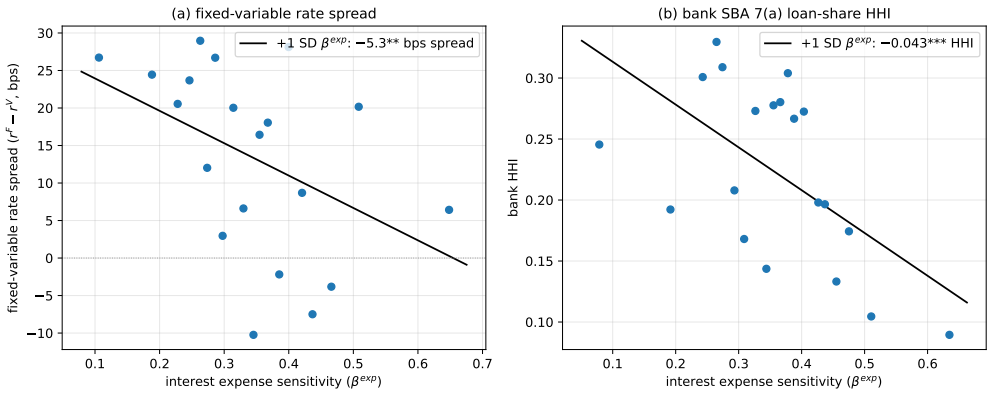


Figure 1: Bank funding sensitivity, fixed-rate premia and lending concentration

Notes: Panel (a) plots the fixed–variable origination-rate differential against bank interest expense sensitivity. Panel (b) plots bank-level SBA 7(a) loan-share HHI against the same sensitivity measure. Dots represent binned observations, and solid lines show fitted relationships. Annotations report the change in each outcome associated with a one-standard-deviation increase in interest expense sensitivity. The loan-share HHI in Panel (b) differs from the lagged branch-count HHI used in the pricing regressions. Sources: FFIEC 031 Call Reports and FR Y-9C consolidated financial statements for bank holding companies, and the Small Business Administration (SBA) 7(a) loan programme, 2010Q1–2019Q4.

to differences in the competitive conditions under which banks price small-business loans, potentially affecting their willingness to supply fixed-rate protection.

At the bank level, Figure 1(b) connects lending concentration to funding sensitivity. Banks with less rate-sensitive funding also operate in more concentrated SBA lending markets: a one-standard-deviation increase in interest expense sensitivity is associated with a 0.043 decline in bank-level SBA loan-share HHI. Stable funding thus tends to coincide with greater lending-market power. Banks with lending power may prefer to restrict fixed-rate lending and preserve higher premia, sacrificing some earnings stability to increase expected earnings. This pricing incentive could counteract the hedging incentive associated with stable funding and help explain the negative premium–sensitivity slope in Figure 1(a).

To assess the role of relative market power, we relate loan-level origination rates to banks' funding sensitivity and one-year-lagged branch-count HHI.[4] County-level branch-count concentration is mapped in Figure 7 in the Empirical Appendix. Figure 2(b) shows that branch-count concentration is positively associated with SBA lending concentration, supporting its relevance for the local lending environment. Using branch counts also mitigates a specific simultaneity concern: contemporaneous SBA lending shares reflect banks' pricing and supply decisions together with borrower demand, so loan-share HHI may be jointly determined with the fixed–variable rate differential. Lagged branch-count HHI captures established banking presence without incorporating current lending volumes. We therefore use it as a proxy for lending-market power conditional on funding sensitivity and the other controls. Since branch networks may still be endogenous, we interpret the estimates as conditional

[4]The pricing sample starts in 2010Q1 with SBA rate-type reporting and ends in 2019Q4 to focus on relatively stable policy-rate uncertainty (Bauer et al., 2022).

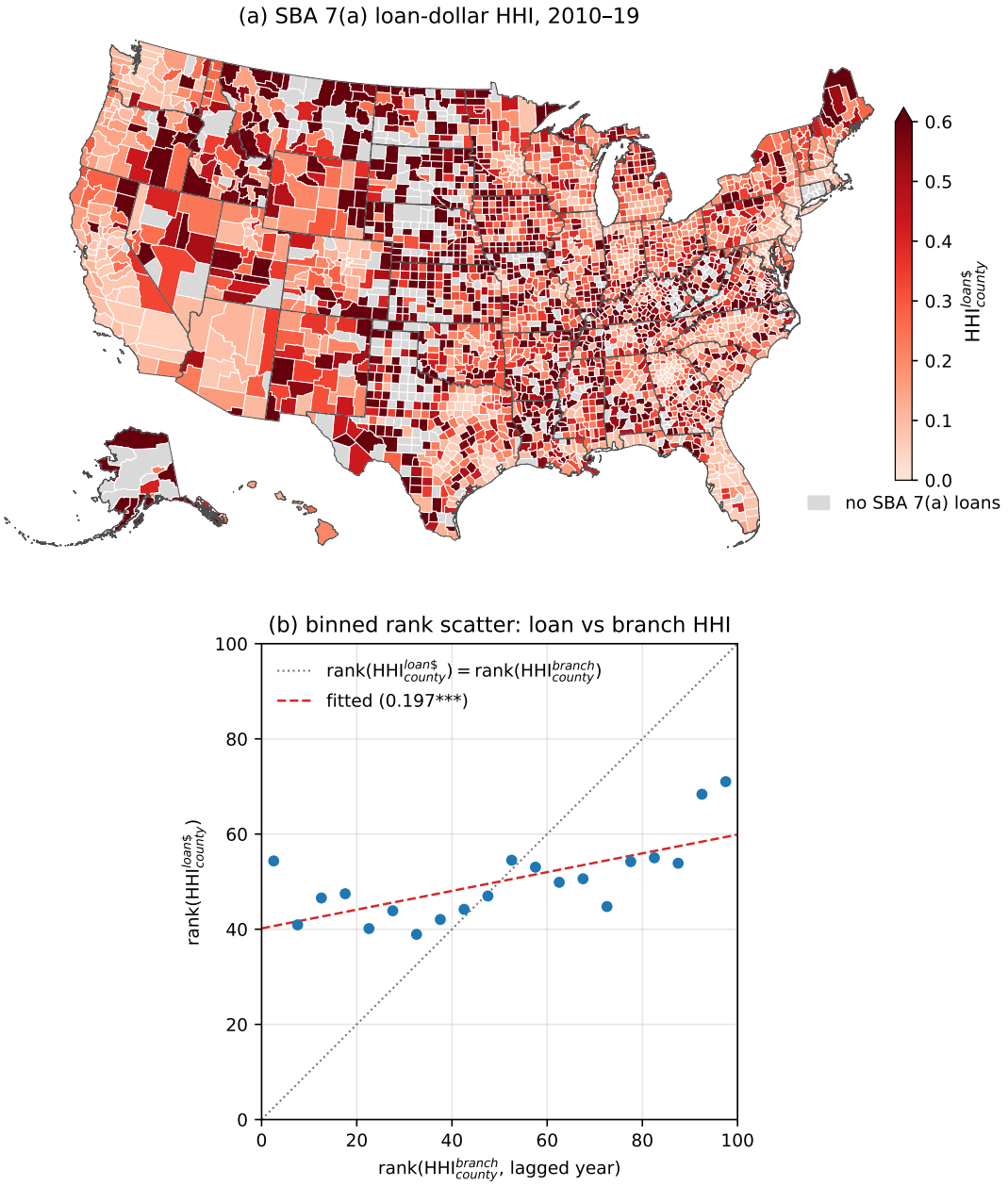


Figure 2: County-level HHI: SBA 7(a) loan share and branch count

Notes: Panel (a) maps county-level concentration measured using banks' shares of SBA 7(a) loan dollars. Grey counties have no SBA 7(a) loans. Panel (b) plots binned county-level ranks of loan-share HHI against ranks of one-year-lagged branch-count HHI. The dashed line shows the fitted relationship, and the dotted line marks equal ranks. Sources: SBA 7(a) loan programme and FDIC Summary of Deposits, 2010–2019.

associations rather than causal effects.

For loan i of rate type $k \in \{F, V\}$, issued by bank b in quarter t, we estimate

$$\begin{aligned} r_{i,k,b,t} = \delta F_i + \alpha_\beta \beta_b^{\exp} + \alpha_H \mathrm{HHI}_{b,t-4} \\ + d_\beta F_i \beta_b^{\exp} + d_H F_i \mathrm{HHI}_{b,t-4} \\ + X_i'\gamma + F_i X_i'\gamma_F + W_{b,t}'\psi + F_i W_{b,t}'\psi_F \\ + \eta_{\text{state},t} + \eta_{\text{industry}} + \varepsilon_i. \end{aligned}$$

Here, $F_i = \mathbf{1}\{k = F\}$ indicates a fixed-rate loan. Loan characteristics X_i include the guaranteed share, jobs supported and maturity, while bank characteristics $W_{b,t}$ include log assets and the equity-to-assets ratio. Their interactions with F_i allow these characteristics to have different associations with fixed- and variable-rate pricing. We include state-by-quarter and borrower-industry fixed effects and cluster standard errors by bank. The interaction coefficients d_β and d_H capture the relationships of funding sensitivity and concentration with the fixed–variable rate differential; the coefficients α_β and α_H capture their relationships with variable-rate pricing.

Table 1: Fixed- and variable-rate loan interest rates, interest expense sensitivity and one-year-lagged branch-count HHI

Dependent variable: interest rate at origination $r_{i,k,b,t}$ (%)	(1)	(2)	(3)	(4)	(5)
	OLS	quarter, industry FE	state×qtr, industry FE	loan-level controls	bank-level controls
Interest expense sensitivity (β_b^{exp})	-0.731	-0.252	-0.503*	-0.462	-0.193
	(0.506)	(0.292)	(0.285)	(0.291)	(0.303)
HHI, lagged one year ($\mathrm{HHI}_{b,t-4}$)	-0.006	-0.044	-0.733	-0.976	0.154
	(1.394)	(0.803)	(0.861)	(0.910)	(0.952)
$F_i\times$ interest expense sensitivity	-0.197	-0.097	0.048	0.656	1.106
	(0.933)	(0.811)	(0.762)	(0.808)	(1.013)
$F_i\times$ HHI (lagged)	1.839	2.452	2.755	3.267*	3.866***
	(2.735)	(1.878)	(1.715)	(1.843)	(1.451)
Loan-level characteristics (X_i, F_iX_i)	No	No	No	Yes	Yes
Bank-level characteristics ($W_{b,t}$, $F_iW_{b,t}$)	No	No	No	No	Yes
Industry FE	No	Yes	Yes	Yes	Yes
State × quarter FE	No	No	Yes	Yes	Yes
Observations	57,018	57,018	57,018	57,018	57,018
Bank-year cells	835	835	835	835	835
Banks	307	307	307	307	307
R^2	0.011	0.382	0.432	0.463	0.481

Notes: standard errors clustered by banks in parentheses; *, **, ***: $p < 0.10$, 0.05, 0.01. Sample: SBA 7(a) Guaranty loans approved 2010Q1–2019Q4 by depository banks who had a branch network in the Summary of Deposits one year before the approval-year snapshot, restricted to bank × calendar-year cells with at least 2 fixed- and 2 variable-rate loans and fewer than 20% of their variable-rate loans at the statutory cap. The interest rate at origination, bank interest expense sensitivity, HHI and loan- and bank-level characteristics are winsorized at the 1st/99th percentiles within rate type. Interest expense sensitivity, $\beta_b^{\exp}$, and bank-level HHI, $\mathrm{HHI}_{b,t}$, are constructed as described in Appendices A.2 and A.3, respectively.

Table 1 reports estimates for 57,018 loans across 835 bank-year observations from 307 banks. The interaction between fixed-rate status and HHI is positive in all five specifications. In the preferred specification, Column (5), its coefficient is 3.866 with a standard error of 1.451, significant at the 1% level. A one-standard-deviation increase in bank lending-market concentration is associated with a 22.5-basis-point higher fixed-rate premium, holding funding sensitivity and the included loan and bank characteristics constant. This finding is consistent with lending-market power supporting higher premia for fixed-rate protection.

The preferred specification also yields a positive interaction between fixed-rate status and funding sensitivity: the coefficient is 1.106 with a standard error of 1.013. Its direction is consistent with the earnings-hedging mechanism. When funding costs are more rate-sensitive, an increase in those costs cannot be matched by repricing fixed-rate loan income, making fixed-rate lending less attractive to banks and increasing the premium they require. The estimate is not statistically distinguishable from zero, however, so the evidence for this conditional funding relationship remains suggestive.

For variable-rate loans, neither funding sensitivity nor concentration has a statistically detectable relationship with pricing in the preferred specification. The corresponding coefficients are -0.193 with a standard error of 0.303 for funding sensitivity and 0.154 with a standard error of 0.952 for HHI. Thus, the clearest pricing association is between concentration and the premium on fixed-rate loans relative to variable-rate loans.

Taken together, these facts highlight the relevance of relative market power in funding and lending for fixed-rate pricing. Stable funding creates an incentive to supply fixed-rate loans as an earnings hedge, while lending-market power creates an incentive to preserve premia by restricting their supply. The concentration result and the direction of the conditional funding-sensitivity estimate are consistent with these opposing influences. They motivate a theoretical model in which banks weigh expected earnings against earnings stability, determining both the fixed-rate premium and the allocation of lending between fixed- and variable-rate contracts.

2.2 Lending composition and relative market power

Having established the relationship between relative market power in funding and lending and the fixed-rate premium, we now turn to loan quantities. We examine how monetary tightening changes the stocks of fixed- and variable-rate loans, and whether the fixed-rate response differs across banks with different lending market power, conditional on their funding sensitivity. We use the monetary policy shocks of Bauer and Swanson (2023), scaled to a contractionary shock of 25 basis points. We use 2002Q2–2024Q4 to exploit more monetary policy variation and improve estimation precision.

Let $l_{b,t}^{k}$ denote the outstanding stock of loans of rate type $k \in \{F, V\}$ held by bank b

in quarter t. We estimate cumulative responses separately by rate type using the following local projection specification:

$$\log l^k_{b,t+h} - \log l^k_{b,t-1} = \phi^k_h \epsilon^{MP}_t + W'_{b,t-1}\Gamma^k_h + Z'_{b,t-1}\Lambda^k_h + \alpha^k_b + \varepsilon^k_{b,t,h},$$

Here, h is the horizon in quarters and ϵ^{MP}_t is the monetary policy shock. The coefficient ϕ^k_h measures the cumulative response of loan stocks from the quarter preceding the shock to horizon h. The vector $W_{b,t-1}$ contains lagged bank characteristics, including asset size, leverage, interest expense sensitivity and HHI. The vector $Z_{b,t-1}$ contains lagged aggregate controls, including the one-year Treasury yield, output growth and the yield-curve slope. We also include bank fixed effects, α^k_b.

Panel (a) of Figure 3 shows that fixed- and variable-rate loan stocks respond differently to monetary tightening. Fixed-rate stocks increase over the first year, reaching approximately 2.6% above their pre-shock level after four quarters, while variable-rate stocks fall by approximately 5.4%. The fixed-rate increase subsequently fades, whereas variable-rate stocks remain below their pre-shock level throughout the remainder of the two-year horizon. Monetary tightening therefore shifts the composition of outstanding lending towards fixed-rate credit.

We next examine whether lending market power is associated with this adjustment. We allow the fixed-rate response to differ across quintiles of bank-level HHI, our proxy for lending market power. We also interact the monetary policy shock with interest expense sensitivity, allowing banks' responses to vary with their funding characteristics. This specification compares the responses across lending-market concentration groups after accounting for differential responses associated with funding sensitivity:

$$\begin{aligned}\log l^F_{b,t+h} - \log l^F_{b,t-1} = &\sum_{q=1}^{5} \phi^F_{h,q} \epsilon^{MP}_t \times 1[\text{quintile}(b) = q] + \theta_h \epsilon^{MP}_t \beta^{\text{exp}}_b \\ &+ W'_{b,t-1}\Gamma^F_h + Z'_{b,t-1}\Lambda^F_h + \alpha^F_b + \varepsilon^F_{b,t,h},\end{aligned}$$

Here, $1[\text{quintile}(b) = q]$ indicates bank b's HHI quintile. The quintile-specific coefficients, $\phi^F_{h,q}$, capture the fixed-rate response within each HHI group, while θ_h captures how the response varies with interest expense sensitivity. The remaining controls and bank fixed effects follow the first specification.

Panel (b) of Figure 3 shows that banks with greater lending market power exhibit the stronger increase in fixed-rate loan stocks. After four quarters, the estimated increase is approximately 4.2% at high-lending-power banks, compared with 0.8% at low-lending-power banks. The high-power response remains larger over subsequent quarters. Conditional on funding sensitivity, the expansion of fixed-rate lending following monetary tightening is therefore more pronounced among banks with greater lending market power.

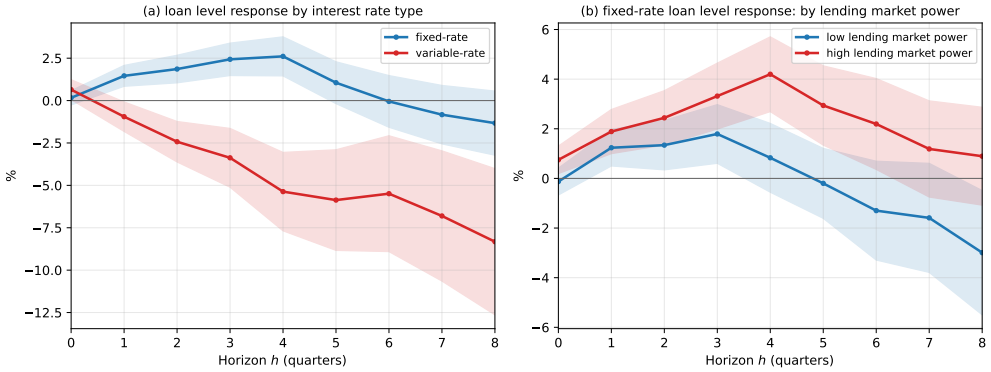


Figure 3: Fixed- and variable-rate lending following +25bps monetary tightening

Notes: Local-projection responses to a contractionary monetary policy shock from Bauer and Swanson (2023), scaled to +25 basis points. Panel (a) compares fixed- and variable-rate loan stocks; Panel (b) compares fixed-rate loan stocks at banks with low and high lending market power, measured by bank-level HHI. Responses are expressed as 100 times cumulative log changes and can be read approximately as percentage changes in loan stocks. Horizons are in quarters, and shaded bands are 68% confidence intervals. Sources: FFIEC 031 Call Reports and FR Y-9C consolidated financial statements for bank holding companies, 2002Q2–2024Q4.

Together, the pricing and quantity evidence connects relative market power to both the pricing of fixed-rate premiums and banks' adjustment of their loan portfolios. Section 3 examines how banks' tradeoff between lending rents and earnings hedging determines fixed-rate premiums and contract choice. Section 4 takes the pricing mechanism to an equilibrium model. By allowing borrowing and investment adjustment, we assess how this mechanism shapes both the composition of lending and the response of overall credit provision to higher interest rates.

3 Model

In this two-date model, stable deposit costs give banks a reason to offer fixed-rate loans at a low premium: when funding costs move little with rates, fixed-rate income helps stabilise earnings. Lending power creates an opposing incentive. To attract more fixed-rate borrowers, a bank must lower the premium on its other fixed-rate loans. The portfolio that best hedges earnings can therefore differ from the one that maximises expected profits. We first hold bank counts fixed, then use entry to explain why deposit and lending competition move together.

3.1 Setup

Loans originate at $t = 0$ and pay at $t = 1$. The only aggregate uncertainty is the net benchmark rate over the life of the loan,

$$r(z) = r_0 + \sigma z, \qquad \Pr(z = 1) = \Pr(z = -1) = \tfrac{1}{2}, \qquad \sigma > 0. \tag{1}$$

Write $x(z)$ for the outcome in state z. Expectations average over the two states: $\mathbb{E}[x(z)] = [x(-1) + x(1)]/2$. Thus $r_0 = \mathbb{E}[r(z)]$ is the mean benchmark rate known at origination, and $\mathrm{Var}(r(z)) = \sigma^2$. All contracts are chosen before z is realised. The parameters and distributions below are common knowledge.

Entrepreneurs. A unit mass of entrepreneurs each borrows one unit to finance a project with deterministic output, fixing total lending at one. Output net of principal repayment, y, ensures participation and repayment in both states. Borrowers choose between fixed and variable rates. They receive no deposit income and have no other hedging instruments.

Entrepreneurs have mean–variance preferences over terminal wealth and heterogeneous risk aversion, $\tilde{\theta} \sim U[0, 2\theta]$, with $\theta > 0$. Each knows its type. A fixed-rate loan charges $r_F = r_0 + s$, where s is the premium set at origination; a variable-rate loan charges $r_V(z) = r(z)$. The pricing asymmetry reflects bank market power in fixed-rate loan market relative to variable-rate loan market.[5] Their certainty equivalents are

$$CE_F = y - r_0 - s, \qquad CE_V(\tilde{\theta}) = y - r_0 - \frac{\tilde{\theta}}{2}\sigma^2.$$

Borrowers choose a fixed rate when $CE_F > CE_V$, or equivalently, $\tilde{\theta} \geq \theta^* \equiv 2s/\sigma^2$. Rate-type sorting based on risk aversion resembles Campbell and Cocco (2003). For an interior allocation, fixed-rate demand and its inverse are

$$L_F = 1 - \frac{s}{\theta\sigma^2}, \qquad s(L_F, \sigma) = \theta\sigma^2(1 - L_F), \qquad \theta^* = 2\theta(1 - L_F). \tag{2}$$

Demand is one for $s \leq 0$ and zero for $s \geq \theta\sigma^2$; equilibrium will be interior. Expanding fixed-rate lending requires attracting less risk-averse borrowers, who will buy protection only at a lower premium. Throughout, $s = r_F - \mathbb{E}[r_V(z)]$ denotes the expected fixed-minus-variable premium.

Deposit and lending markets. There are $M \geq 1$ local deposit markets and a common lending market. With integer bank counts, each local market has n banks and the total

[5]The protection offered by fixed-rate financing has few accessible substitutes: Vickery (2008) documents financially constrained small firms disproportionately choose fixed-rate loans and seldom use derivatives to hedge interest rate risk. Public bond market is often costly to access for smaller firms (Faulkender, 2005).

is $N = Mn$. We allow $n \geq 1$ and $N = Mn$ to vary continuously as effective competition indices: n determines access to competing deposit offers, while $1/N$ measures an individual lender's influence on the fixed-rate premium. Integer values admit the symmetric bank-count interpretation; non-integer values use its continuous extension. Depositors compare offers locally, whereas borrowers access the common lending market. We initially treat n and M as primitives. Section 4 allows heterogeneous local markets and segments lending as well as deposits.

Total deposits are one, divided equally across banks in the integer benchmark, each with predetermined loan capacity $d_i = 1/N$. Depositors are risk-neutral and have an outside savings return $\underline{r} > 0$. This exogenous reservation rate is known at origination and fixed across interest-rate states, competition indices, and market sizes. Its provider and pricing are outside the model. Households cannot directly access the wholesale benchmark $r(z)$. We assume $r_0 > \underline{r}$: depositors compare expected returns, so the benchmark need not exceed $\underline{r}$ in every state. We focus on parameter configurations in which banks do not voluntarily increase deposit pass-through to hedge earnings. Appendix D.3 gives sufficient conditions and their compatibility with a negative premium–beta relationship.

Deposit search and pricing. Each depositor has an offer from its incumbent and compares it with a rival's offer if one arrives before the decision deadline. Measure search time in units that give the deadline an exponential arrival rate of one. Each of the $n - 1$ rivals makes contact independently at rate $\zeta > 0$ in these units. Thus ζ measures the contact rate per rival relative to the deadline rate. With a continuum of accounts, the fraction receiving a competing offer is

$$p(n; \zeta) = \frac{\zeta(n-1)}{1 + \zeta(n-1)}. \tag{3}$$

Throughout this section we assume equal contact and deadline rates, $\zeta = 1$, and write $p(n) = p(n; 1) = 1 - 1/n$. Search is exogenous and independent of z and borrower types. The aggregate contact rate extends to real $n \geq 1$; the constant market-breadth threshold below uses $\zeta = 1$.

Eligible banks know whether a depositor has a competing offer and quote personalised indexed contracts $\underline{r} + \alpha(r(z) - \underline{r})$, with $\alpha \geq 0$. Here α is account-level passthrough: $\alpha = 0$ pays $\underline{r}$, while $\alpha = 1$ pays $r(z)$. Since $r_0 > \underline{r}$, risk-neutral depositors prefer larger α and remain with the incumbent when expected returns tie, including a tie with outside savings. Quotation is costless and separate fees are excluded. Banks can replace a lost deposit by borrowing at $r(z)$, and invest a newly acquired surplus deposit at $r(z)$. This wholesale market accommodates deposit movements against committed loan books; households cannot access it directly.

Competition bids contested accounts up to $r(z)$, eliminating their funding advantage rel-

ative to wholesale borrowing. Switching at this rate would leave the incumbent's funding cost unchanged, while the receiving bank could invest the extra deposit at the same rate. In the maintained regime, uncontested accounts receive $\underline{r}$. The tie-breaking rule keeps deposits with their incumbents: competition changes rates even without realised deposit flows. Appendix D.2 verifies optimal bids, allowing for their hedging benefits; $r(z)$ is an equilibrium bidding limit, not an imposed cap. Equilibrium average funding costs and passthrough are

$$r^d(z) = (1-\beta)\underline{r} + \beta r(z), \qquad \beta \equiv \frac{\partial r^d(z)}{\partial r(z)} = p(n) = 1 - \frac{1}{n}.$$

More local competitors expose more accounts to rival offers, making funding costs more responsive to rates. As $n \to \infty$, banks pass through rate changes one-for-one, i.e. $\beta \to 1$. In contrast, when there is a monopoly bank, i.e. $n = 1$, the bank does not pass through any rate changes, i.e. $\beta = 0$. The exogenous $\underline{r}$ sets the funding-cost floor without directly changing pass-through; a rate-sensitive outside return would change this conclusion. The construction uses personalised bids and observable comparison sets, rather than a Burdett–Judd posted-rate equilibrium.

Lending and bank preferences. In the integer benchmark, bank i chooses its fixed-rate portfolio share $l_{F,i} \in [0,1]$. Its fixed quantity and aggregate fixed lending are

$$q_{F,i} = d_i l_{F,i} = \frac{l_{F,i}}{N}, \qquad L_F = \sum_{i=1}^{N} q_{F,i} = \frac{1}{N}\sum_{i=1}^{N} l_{F,i}.$$

We focus on symmetric equilibria and suppress the bank index i in the remainder of this section. The remainder of its book is lent at the variable rate. Banks commit to fixed quantities and compete in Cournot fashion, internalising $\partial s/\partial l_F = -\theta\sigma^2/N$. The continuous extension retains this price-impact coefficient in the symmetric pricing conditions. All borrowers face the same fixed premium, while banks take $r_V(z) = r(z)$ as given. Borrower types can be allocated across banks to fill their capacities.

With retained deposits and bank-specific average pass-through β, net interest earnings per unit of assets are $\pi(z) = l_F r_F + (1-l_F)r_V(z) - r^d(z)$. Substituting the contract rates gives

$$\pi(z) = (1-\beta)(r_0 - \underline{r}) + l_F s(L_F, \sigma) + (1-\beta-l_F)\sigma z. \tag{4}$$

Principal repayments cancel. The sensitivity gap is

$$G \equiv \frac{\partial \pi(z)}{\partial r(z)} = (1-l_F) - \beta, \tag{5}$$

so earnings variance is $\sigma^2 G^2$. The gap captures the difference between interest income and

interest expense sensitivities. A positive gap means that higher rates raise bank earnings; a negative gap means that they lower earnings. Banks maximise

$$\max_{l_F \in [0,1]} \left\{ \mathbb{E}\pi(z) - \frac{\gamma}{2} \operatorname{Var}(\pi(z)) \right\}.$$

Throughout, γ applies to earnings per unit of assets (Hart & Jaffee, 1974). Credit losses, capital requirements, prepayment, separate origination fees, and additional hedging instruments are excluded.

Timing. Comparison sets are realised and observed by eligible banks before the rate shock. Banks then choose fixed-loan quantities, anticipating the deposit continuation. The fixed premium clears the loan market and entrepreneurs select contracts. Eligible banks subsequently quote indexed deposit contracts, depositors choose their accounts, and wholesale funding accommodates any movement. All contracts settle after z is realised. In the selected continuation all deposits are retained and wholesale positions are zero.

3.2 Equilibrium

For given integer counts n, N, banks choose portfolios taking rivals' quantities as given and anticipating optimal deposit bids after every feasible portfolio profile. Borrowers and depositors choose their preferred available contracts, and loan and funding balances clear. We select the deposit retention continuation above and focus on symmetric lending, with $l_F = L_F$ and $G = 1 - \beta - L_F$. For non-integer competition indices, the corresponding symmetric conditions define the continuous extension.

Lemma 1 (Fixed-rate pricing in symmetric Cournot equilibrium)**.** *Fix n and N and suppose the deposit-retention continuation yields $\beta = 1 - 1/n$ at equilibrium. In the symmetric Cournot equilibrium, the fixed-rate premium and the common fixed-loan share $l_F = L_F$ satisfy*

$$s = \underbrace{\frac{\theta \sigma^2 L_F}{N}}_{\textit{Cournot term}} - \underbrace{\gamma \sigma^2 G}_{\textit{marginal hedging benefit}}. \tag{6}$$

Proof. See Appendix C.1. □

When funding costs respond less to rates than loan income ($G > 0$), replacing variable-rate loans with fixed-rate loans stabilises earnings. The bank will accept a lower premium for this hedge. Yet attracting another borrower requires a lower premium on all the fixed loans it originates. The Cournot term measures the premium revenue forgone on its other loans, which the bank weighs against the additional loan's income and hedging benefit.

Banks choose fixed quantities at origination, when the common premium clears the market. Existing fixed loans retain their contracted rate.

With lending power, the bank can therefore prefer to supply less protection and tolerate earnings fluctuations in order to earn higher expected profits. Stable funding gives it a reason to supply protection cheaply, while lending power rewards restricting that supply. This hedging argument applies while $G > 0$: once fixed lending exceeds the full hedge, $G < 0$ and additional protection increases the bank's earnings risk, requiring compensation. The exogenous $\underline{r}$ affects expected funding costs but, conditional on bank counts, neither marginal incentive in (6).

Proposition 1 (Equilibrium and earnings exposure)**.** *Conditional on the deposit retention equilibrium, there is a unique symmetric lending equilibrium. Let $\beta = 1 - 1/n$. Then*

$$L_F = \frac{\theta + \gamma(1-\beta)}{\theta(1+1/N)+\gamma} = \frac{\theta + \gamma/n}{\theta(1+1/N)+\gamma}. \tag{7}$$

The corresponding premium is

$$s = \theta\sigma^2 \frac{\theta/N + \gamma\beta}{\theta(1+1/N)+\gamma}, \tag{8}$$

and earnings exposure is $G = \frac{\theta[1-(N+1)\beta]}{N[\theta(1+1/N)+\gamma]}$. The allocation is interior, with $0 < L_F < 1$ and $0 < s < \theta\sigma^2$. The sensitivity gap is positive, zero, or negative as β is below, equal to, or above $1/(N+1)$.

Proof. Combine (2) and (6) to obtain $[\theta(1+1/N)+\gamma]L_F = \theta+\gamma(1-\beta) > 0$. The denominator exceeds the numerator by $\theta/N + \gamma\beta > 0$, establishing interiority. Strict own concavity makes the first-order condition sufficient. The symmetric linear equation has one solution, while the symmetric all-zero and all-one portfolios fail the boundary optimality conditions. Substitution gives the premium and the gap. □

Two portfolio targets explain the allocation. Full earnings hedging requires $L^{hedge} = 1 - \beta = 1/n$, while a risk-neutral bank chooses $L^{Cournot} = N/(N+1)$. Equilibrium satisfies

$$L_F = \omega L^{hedge} + (1-\omega)L^{Cournot}, \qquad \omega = \frac{\gamma}{\theta(1+1/N)+\gamma} \in (0,1).$$

When $L^{hedge} > L^{Cournot}$, preserving lending rents leads banks to stop short of the full hedge: they retain $G > 0$ and forgo some profit stability. When $L^{hedge} < L^{Cournot}$, earning premium income instead draws them beyond the full hedge, leaving $G < 0$. In both cases, banks accept earnings risk to raise expected profits relative to the perfectly hedged portfolio. Greater risk aversion raises the weight on L^{hedge}, but complete hedging occurs only when the two targets coincide.

Greater rate uncertainty σ raises the premium but leaves the fixed share unchanged. Under mean–variance preferences, borrowers' willingness to pay and banks' compensation for bearing risk both scale with variance. Other preferences or endogenous participation (Section 3.6) can change the share response.

3.3 Competition and the premium–beta relationship

Funding and lending competition change different sides of this tradeoff. More responsive funding reduces the fixed-rate share needed to stabilise earnings, weakening banks' willingness to supply borrower protection. More lenders instead reduce each bank's incentive to restrict fixed lending to preserve the premium.

First vary funding and lending competition separately. Although deposit passthrough is determined by local competition, $\beta = 1 - 1/n$, we first treat it as exogenous for ease of exposition. Holding the other margin fixed gives

$$\left.\frac{\partial s}{\partial \beta}\right|_N = \frac{\theta\gamma\sigma^2}{\theta(1+1/N)+\gamma} > 0, \qquad \left.\frac{\partial s}{\partial N}\right|_\beta = -\frac{\theta^2\sigma^2[\theta+\gamma(1-\beta)]}{N^2[\theta(1+1/N)+\gamma]^2} < 0.$$

To study their joint effect, hold market breadth $M = N/n \geq 1$, the number of local deposit markets accessible to borrowers, fixed and vary n. In the maintained deposit-pricing regime, increasing n raises both $\beta = 1 - 1/n$ and $N = Mn$. Since $N = M/(1-\beta)$, the premium–beta slope decomposes into

$$\left.\frac{ds}{d\beta}\right|_M = \underbrace{\left.\frac{\partial s}{\partial \beta}\right|_N}_{\text{funding effect } (>0)} + \underbrace{\left.\frac{\partial s}{\partial N}\right|_\beta \frac{M}{(1-\beta)^2}}_{\text{lending-competition effect } (<0)}.$$

Therefore, *a negative premium–beta slope arises when the lending-competition effect dominates the funding effect.* The following proposition establishes the condition on market breadth.

Proposition 2 (Market breadth and the premium slope)**.** *Along symmetric equilibria with fixed $M = N/n$ and deposit passthrough $\beta = 1 - 1/n$, the fixed-rate premium decreases with deposit beta if*

$$M < \frac{\theta^2}{\gamma(\theta+\gamma)},$$

increases if the inequality is reversed, and remains constant at equality. Under the uniform sufficient condition in Appendix D.3, the result holds over $\beta \in [0, 1)$ and orders the corresponding integer-count configurations.

Proof. Writing $\rho = \gamma/\theta$, substitution into (8) gives

$$\frac{s(\beta; M)}{\theta\sigma^2} = \frac{1 + (\rho M - 1)\beta}{1 + M(1+\rho) - \beta}. \tag{9}$$

Differentiating (9) at fixed M yields

$$\frac{ds}{d\beta} = \theta\sigma^2 \frac{M[\rho M(1+\rho) - 1]}{[1 + M(1+\rho) - \beta]^2}.$$

The denominator is positive, giving the result. □

When borrowers rely on a narrow lending market, entry can substantially reduce the rents banks preserve by restricting fixed lending. The resulting expansion of protection can outweigh the loss of the hedging incentive as funding becomes more rate-sensitive. When borrowers already access many lenders, there is less lending power left to erode, so the funding effect dominates. A negative slope is feasible for $M \geq 1$ only if $\gamma/\theta < (\sqrt{5} - 1)/2$. Figure 4 illustrates both regimes; the premium itself remains positive.

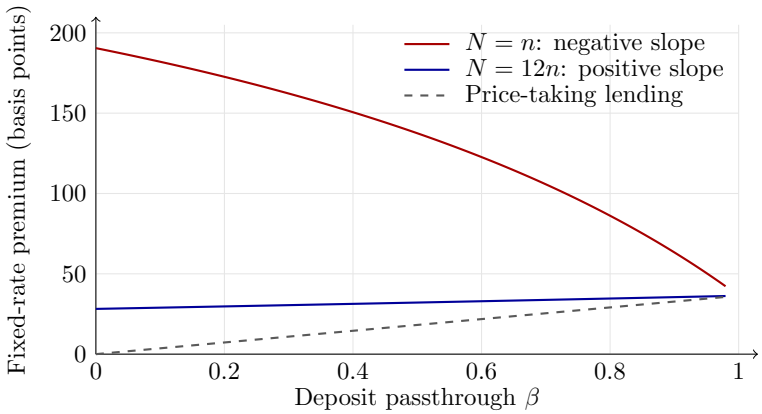


Figure 4: Fixed-rate premiums and deposit pass-through. The illustration uses $\theta = 1600$, $\gamma = 160$, $\sigma = 0.005$, and $r_0 - \underline{r} = 0.03$, giving a breadth threshold of 9.09 and satisfying the deposit-pricing condition in Appendix D.3. Solid curves hold N/n fixed and vary the continuous competition indices; the dashed curve takes $N \to \infty$ at a fixed beta. The curves are theoretical illustrations, not estimates.

Price-taking lending. Let $N \to \infty$ while holding n, and hence deposit passthrough β, fixed. This case has no lending market power. The Cournot term in (6) disappears and (8) implies

$$s_\infty(\beta) = \frac{\theta\gamma\sigma^2}{\theta + \gamma}\beta.$$

With completely rate-insensitive funding, banks can supply an entirely fixed-rate loan book without exposing earnings to interest-rate risk, so the premium is zero. For positive beta, the

competitive allocation takes fixed lending beyond the share that fully hedges bank earnings. The positive premium compensates banks for the additional earnings risk they bear when providing borrower protection.

Taking $n \to \infty$ at fixed M would instead increase competition in both markets and send deposit beta to one. In the limit considered here, deposit rents can survive price-taking lending because local funding competition remains unchanged and banks choose the composition of predetermined loan books.

3.4 The premium–beta relationship under CARA and CRRA

The fixed-rate premium can fall as deposit passthrough rises under both constant absolute and constant relative risk aversion. As in Section 3.3, more responsive deposit costs weaken a bank's incentive to hedge with fixed-rate loans. But when borrowers gain access to more lenders, each bank also has less power to sustain a high premium. To establish when lending competition dominates, keep the same deposit contracts and borrower types and let borrowers and banks maximise expected utility.

With a slight abuse of notation, retain $\tilde{\theta} \sim U[0, 2\theta]$ for borrower risk aversion and γ for bank risk aversion. For a borrower set $\varrho = \tilde{\theta}$ and for a bank set $\varrho = \gamma$. Consider two utility functions over terminal wealth w:

$$u_\varrho^{\text{CARA}}(w) = \frac{1 - e^{-\varrho w}}{\varrho}, \qquad u_\varrho^{\text{CRRA}}(w) = \begin{cases} (w^{1-\varrho} - 1)/(1-\varrho), & \varrho \neq 1, \\ \log w, & \varrho = 1. \end{cases}$$

Under constant absolute risk aversion (CARA), ϱ measures aversion to a given dollar risk. Under constant relative risk aversion (CRRA), it measures aversion to a given proportional risk, and w must be positive. The parameters therefore have different units across specifications. In either case $\varrho = 0$ gives linear utility, taking the limit for CARA.

For any given aggregate fixed-rate share L, the most risk-averse fraction L of borrowers choose fixed loans. The indifferent borrower therefore has type $2\theta(1-L)$. Let $c = y - r_0 > \sigma$ be its consumption after paying the expected variable rate. Inverse demand $s(L, \sigma)$ is the premium at which this borrower is indifferent between consuming $c - s$ for certain and consuming $c - \sigma$ or $c + \sigma$ with equal probability.

For both preferences, let the bank start with the same predetermined equity E per unit of assets and evaluate terminal equity $E + \pi(z)$. At a symmetric allocation its earnings exposure is $G = 1 - \beta - L$, and its terminal equity per unit of assets is $W \pm \sigma G$, where $W = E + (1-\beta)(r_0 - \underline{r}) + Ls$. Under CARA, adding the same equity in both states leaves their relative marginal utilities unchanged. Under CRRA, a given earnings fluctuation matters more when equity is low. We choose E so that terminal equity is positive in both states for every feasible portfolio; $E = 0$ also suffices if earnings alone meet this condition.

Lemma 2 (Fixed-rate pricing under expected utility)**.** *In an interior symmetric Cournot equilibrium, the fixed-rate premium satisfies*

$$s = \underbrace{-\frac{L}{N}\frac{\partial s(L,\sigma)}{\partial L}}_{\text{Cournot term}} - \underbrace{H}_{\text{marginal hedging benefit}}, \qquad H = \sigma\frac{u'_\gamma(W-\sigma G) - u'_\gamma(W+\sigma G)}{u'_\gamma(W-\sigma G) + u'_\gamma(W+\sigma G)}, \tag{10}$$

where u_γ is the bank's utility under the specification considered.

Proof. See Appendix C.2. □

Consider a risk-neutral bank taking rivals' quantities as given. An additional fixed loan earns the premium s but lowers the premium on the bank's other fixed loans. Since $\partial L/\partial l_F = 1/N$, its first-order condition is $s+(l_F/N)s_L = 0$. At the common equilibrium share $L^{Cournot}$, this becomes

$$s(L^{Cournot},\sigma) + \frac{L^{Cournot}}{N}s_L(L^{Cournot},\sigma) = 0. \tag{11}$$

Now consider a risk-averse bank at that same portfolio. If $L^{Cournot} < 1-\beta$, its earnings are lower in the low-rate state. An additional fixed loan shifts income into that state. Its marginal expected profit is zero at $L^{Cournot}$, but its hedging benefit is positive, so the bank wants to expand fixed lending. If $L^{Cournot} > 1-\beta$, the same loan adds to earnings risk and the bank wants to supply less. Under CARA the earnings gap determines the value of this hedge. Under CRRA a bank with less equity values the same reduction in risk more.

Proposition 3 (Portfolio choice under expected utility)**.** *Let $L^{Cournot}$ denote the symmetric Cournot fixed share with risk-neutral banks that solves* (11)*, and $L^{hedge} = 1-\beta$ the share that fully hedges bank earnings. Under CARA the symmetric Cournot equilibrium is unique. Its fixed share lies between $L^{Cournot}$ and L^{hedge}, strictly so when they differ. The same result holds under CRRA if premium revenue is strictly concave in the bank's own fixed share, symmetric marginal revenue is strictly decreasing in L, and G/W decreases in L.*

Proof. See Appendix C.2. □

To compare markets, again hold $M = N/n$ fixed. As deposit beta rises, the bank needs fewer fixed loans to hedge deposits. At the same time, borrowers gain access to more lenders, reducing the premium revenue each bank gives up when it expands fixed lending.

Proposition 4 (Market breadth and the premium slope under expected utility)**.** *Along $N = Mn$, suppose the marginal return $s+(L/N)s_L+H$ is strictly downward sloping in L at the symmetric equilibrium. The fixed-rate premium decreases with deposit beta if and only if*

$$\frac{L(-s_L)}{M} > -\left.\frac{\partial H}{\partial \beta}\right|_L.$$

Under the conditions of Proposition 3, if symmetric marginal revenue has strictly negative slope at $L^{Cournot}$, this inequality holds for sufficiently small positive γ under either utility function.

Proof. See Appendix C.2. □

The left side measures how much entry weakens the incentive to restrict fixed lending. The right side measures how much more responsive deposits weaken the incentive to hedge. When banks place little weight on earnings risk, the pricing incentive dominates even though banks still value a hedge.

In the CRRA illustration, doubling n from two to four raises deposit beta from 0.50 to 0.75. Only market breadth differs across economies. Set $\theta = 2$ (borrower types range from zero to four), $\gamma = 0.80$, $c = 0.08$, $E = 0.10$, and $r_0 - \underline{r} = 0.03$. A larger shock, $\sigma = 0.02$, makes the premium changes easier to see.

With two deposit markets ($M = 2$), borrowers gain access to eight lenders instead of four ($N = Mn$). The premium falls from 25.66 to 24.50 basis points and the fixed share rises from 74.75% to 75.89%. With twelve deposit markets ($M = 12$), borrowers already face 24 lenders before the increase and 48 afterwards. The premium rises from 13.73 to 18.45 basis points and the fixed share falls from 86.49% to 81.86%.

Where borrowers initially have few lenders, competition makes protection cheaper despite weaker hedging incentives. In a broad market, lenders have less pricing power to lose and the funding effect can raise the premium.

3.5 Welfare and the allocation of interest-rate risk

Deposit competition determines how closely funding costs follow market rates. A bank with stable deposit costs can insure borrowers against a rate increase at little risk to its earnings. Lending competition determines how much of that protection reaches borrowers: selling another fixed loan requires a bank with pricing power to lower the premium on its other fixed loans.

Compare efficient and Cournot lending for the same deposit contracts, bank counts, loan menu, and total credit. Premium payments transfer income from borrowers to banks. The loss arises when a borrower retains rate risk even though fixing the loan would raise joint surplus.

Let L be the share of loans allocated to fixed-rate borrowers, starting with the most risk averse. Borrowers with $\tilde{\theta} < 2\theta(1 - L)$ retain variable-rate loans, and each bank retains an earnings exposure of $1 - \beta - L$. Summing borrower and depositor certainty equivalents and weighting banks' per-asset certainty equivalents by their asset shares gives, up to a constant,

$$\mathcal{W}(L) = -\frac{\sigma^2}{2}\left[\theta(1 - L)^2 + \gamma(1 - \beta - L)^2\right]. \tag{12}$$

The first term measures the cost of rate risk borne by borrowers and the second the cost borne by banks. Risk-neutral depositors require no compensation for risk. The efficient share,

$$L^\star = \frac{\theta + \gamma(1-\beta)}{\theta + \gamma},$$

gives more weight to insuring borrowers when θ is high and to hedging bank earnings when γ is high. At the Cournot allocation, $\mathcal{W}'(L_F) = \theta\sigma^2 L_F/N > 0$: the gain from one more fixed loan equals the premium revenue the bank would forgo on its other fixed loans. By (12), the loss from this shortfall is

$$\mathcal{W}(L^\star) - \mathcal{W}(L_F) = \frac{\sigma^2}{2}(\theta+\gamma)(L^\star - L_F)^2.$$

Under mean–variance preferences, greater uncertainty raises both parties' risk costs in proportion to σ^2. The efficient and equilibrium shares therefore stay fixed, and doubling σ quadruples the welfare loss for any γ/θ.

Uncertainty and risk aversion. With CARA preferences, a more risk-averse borrower will pay more to fix the rate. What makes it profitable for a bank to withhold a loan, however, is the difference between what it can charge its current borrowers and what it must charge to attract the next one. For small shocks, marginal willingness to pay is approximately $\theta\sigma^2(1-L)$. A higher θ then steepens inverse demand: expanding fixed lending requires a larger premium cut on the bank's existing fixed volume. This is one way borrower risk aversion can make lending power more costly.

Proposition 5 (Uncertainty and the welfare cost of lending power)**.** *Under CARA preferences with $\theta, \gamma > 0$, fix deposit pass-through $0 < \beta < 1$ and a finite number of lenders. Let $\Delta(\sigma)$ be the gain in aggregate monetary certainty equivalents from replacing symmetric Cournot lending with efficient lending. The gain is positive for every $\sigma > 0$ and attains its maximum at an interior level of uncertainty. It tends to zero as $\sigma \to 0$ and as $\sigma \to \infty$.*

Proof. See Appendix C.3. □

The loss therefore has an interior peak. At low uncertainty, an increase in σ makes each loan left unhedged more costly. At high uncertainty, even the marginal fixed-rate borrower will pay close to σ for protection. Borrowers' valuations converge relative to the size of the shock, while banks place increasing weight on losses in their adverse state. Both allocations approach the bank's full hedge, $1-\beta$, and the additional loss due to lending power vanishes. This large-shock comparison holds deposit contracts fixed.

For $n = N = 2$ and $\beta = 1/2$, raise σ from 0.005 to 0.015. At $\theta = 1600$, the loss rises from 2.74 to 4.14 basis points per unit lent for $\gamma = 80$ but falls from 1.24 to 0.46 for $\gamma = 320$.

Borrower preferences can also reverse the comparison. At $\gamma = 160$, the loss rises from 2.08 to 2.12 basis points for $\theta = 800$ but falls from 2.11 to 1.91 for $\theta = 1600$, even though these borrowers value protection more.

3.6 Market size and endogenous entry

Section 3.3 shows when the premium falls as bank counts rise. What makes depositors and borrowers face more competitors in the first place? A larger customer base can cover the fixed cost of another bank. Because each entrant takes deposits locally and lends across communities, one participation decision expands choice on both sides of the market.

Let $\mathcal{A} > 0$ denote both the mass of unit loans and total deposits. Each of the M communities supplies $\mathcal{A}/M$ deposits. Search remains uncongested, so market size affects passthrough only through entry. For this extension assume $r_0 - \underline{r} > \gamma\sigma^2$, which rules out voluntary extra passthrough after any portfolio choice and supports the franchise-value inequalities below. Earlier results require only the weaker condition in Appendix D.3.

To operate, owners pay $F > 0$ in terminal goods from outside resources. This cost is independent of market size and beta. Staying out pays zero. Before search, active banks share home-community deposits equally, extending the benchmark allocation without a separate franchise acquisition price. We study configurations with at least one bank per community.

Under symmetry, $N = Mn$ banks each hold $\mathcal{A}/(Mn)$ in assets. Fixed volume is $\mathcal{A}L_F$, so inverse demand (2) and all per-asset outcomes carry over. Risk-adjusted value per unit of assets before entry costs is

$$W(n; M) = \frac{r_0 - \underline{r}}{n} + L_F(n; M)s(n; M) - \frac{\gamma}{2}\sigma^2 \left[\frac{1}{n} - L_F(n; M)\right]^2, \tag{13}$$

and $v(n; M) = W(n; M)/(Mn)$ gives bank value per unit of market size. Entry yields $\mathcal{A}v(n; M) - F$. At given counts a higher outside return leaves portfolios unchanged but erodes deposit rents, $\partial W/\partial \underline{r} = -1/n$. We hold $\underline{r}$ fixed.

We use a continuous bank count to trace how participation responds to market size. Appendix D.5 gives the exact entry conditions for integer counts in one or several communities.

Proposition 6 (Market size and participation)**.** *Under the maintained parameter restrictions, $W(n; M) > 0$ and $W_n(n; M) < 0$ for every real $n \geq 1$ and $M \geq 1$. Consequently $v(n; M)$ is strictly decreasing and tends to zero.*

For any fixed $M \geq 1$, the continuous symmetric participation equation

$$\frac{\mathcal{A}}{Mn} W(n; M) = F \tag{14}$$

has a unique solution $n > 1$ whenever $\mathcal{A}W(1; M)/M > F$. Along this approximation,

$$\frac{dn}{d\mathcal{A}} = \frac{n/\mathcal{A}}{1 - nW_n/W} > 0, \qquad \frac{d\beta}{d\mathcal{A}} = \frac{1}{n^2}\frac{dn}{d\mathcal{A}} > 0, \qquad \frac{dN}{d\mathcal{A}} = M\frac{dn}{d\mathcal{A}} > 0. \tag{15}$$

The premium–beta slope is again given by Proposition 2.

Proof. See Appendix D.4. □

Entry divides the larger market among more banks. More depositors receive competing offers and more lenders compete for fixed-rate borrowers. Both margins erode rents, limiting further entry.

At fixed M, market size thus selects a point on the premium–beta curve in Section 3.3. Larger markets have higher beta and cheaper protection when competition for borrowers dominates. When more responsive funding matters more for hedging, beta and the premium rise together. Proposition 2 gives the threshold.

4 Loan pricing, investment, and the macroeconomy

We extend the loan-pricing mechanism to an economy in which entrepreneurs choose project size as well as financing type. A higher fixed-rate premium reduces investment by borrowers who remain fixed and induces others to switch to smaller floating-financed projects. Bank pricing therefore affects both the allocation of interest-rate risk and real activity, even at unchanged expected benchmark rates. Household saving and labour supply determine funding and wage feedback. Banks can also hedge externally, substituting for fixed lending as an earnings hedge while retaining the incentive to restrict credit for lending rents.

Environment. There are M markets. Market m has a mass φ^m of entrepreneurs, with $\sum_{m=1}^{M} \varphi^m = 1$. As in Section 3.6, we use a continuous count n^m for active banks and write $N = \sum_m n^m$. Here each bank takes deposits and lends in its own market. Borrowers access a subset of local lenders, as specified in Section 4.2. Depositors search as in (3), with local contact rate ζ^m relative to the decision deadline. Under the same deposit-pricing rule,

$$r_d^m(z) = (1 - \beta^m)\underline{r} + \beta^m r(z), \qquad \beta^m = p(n^m; \zeta^m). \tag{16}$$

The real interbank rate $r(z)$ follows (1). A government-operated facility implements it by accepting surplus bank funds and lending to deficit banks at the same state-contingent rate; balanced positions require no net fiscal transfer (Appendix E.2). The household outside return $\underline{r}$ is distinct from the mean interbank rate r_0: we hold it fixed in the main rate experiment, while saving and investment remain endogenous.

Entrepreneurs borrow in their assigned markets, while labour moves freely across markets at a common wage. Local credit conditions thus feed into economy-wide wages and employment, with no additional labour-market segmentation. Retailers contract their date-0 supply prices before changes in r_0 or σ and price flexibly at date 1. These inherited contracts allow current output and employment to respond to financing conditions.

Banks observe both r_0 and σ before entry, deposit search and portfolio choices. Loan terms, outside hedges, investment and employment at both dates are chosen at date 0, before z is realised. The terminal wage w_1 clears the market for committed date-1 labour at date 0: it adjusts across equilibria, but not to the subsequent rate realisation. This timing isolates financing risk in loan choice while retaining wage and employment adjustment to changes in banking conditions. Contracts settle at date 1.

4.1 Borrower investment and loan choice

Loan pricing affects real activity because entrepreneurs choose project scale together with the financing contract. Each borrows the full cost of investment k, hires h hours and produces $A_1 k^\alpha h^{1-\alpha}$ intermediate goods, where $A_1 > 0$ is productivity and $\alpha \in (0,1)$ is the capital share. Output sells at $\bar{p} = (\varepsilon - 1)/\varepsilon$, with retail substitution elasticity $\varepsilon > 1$. Capital depreciates fully, and expansion costs $\kappa k^2/2$ terminal goods, with $\kappa > 0$.

In market m, borrower risk aversion satisfies $\log \tilde{\theta}^m \sim N(\mu^m, (\tau^m)^2)$, where μ^m and $\tau^m > 0$ are the market-specific mean and standard deviation of log risk aversion. Fixed finance (F) charges $r_F^m = r_0 + s^m$, with local premium s^m; variable finance (V) charges $r_V(z) = r(z) = r_0 + \sigma z$. The rate-shock states $z \in \{-1, 1\}$ are equally likely, and $\sigma > 0$ measures uncertainty. Given loan rates and the common wage w_1, the entrepreneur chooses contract j, investment and committed hours before z is realised:

$$\max_{j \in \{F,V\},\, k,h \geq 0} -\mathbb{E}\Big[\exp\Big(-\tilde{\theta}^m \Pi_j^m(z)\Big)\Big], \tag{17}$$

Expectations average over the two rate states. The payoff $\Pi_j^m(z)$ is output revenue less wages, loan repayment and expansion costs, including any profit-equivalent contract benefit; dependence on k, h is suppressed. Fixed-rate profit Π_F^m is independent of the state and varies across markets through s^m. For given k, h, variable-rate profit is common across markets, so $\Pi_V^m(z) = \Pi_V(z)$. Optimal labour demand is proportional to project size:

$$h = \left[\frac{(1-\alpha)\bar{p}A_1}{w_1}\right]^{1/\alpha} k.$$

Substituting labour demand gives

$$\Pi_F^m = (R - s^m + v_F)k - \frac{\kappa}{2}k^2, \qquad \Pi_V(z) = (R - \sigma z)k - \frac{\kappa}{2}k^2.$$

Here $v_F k$ is a nonpecuniary convenience benefit, for example from simpler payment administration. Its additive, proportional form assumes constant willingness to pay per unit financed in profit-equivalent units, separate from the insurance value captured by risk aversion. The common expected surplus is

$$R = \alpha \bar{p} A_1 \left[\frac{(1-\alpha)\bar{p}A_1}{w_1}\right]^{(1-\alpha)/\alpha} - 1 - r_0. \tag{18}$$

Thus R is the expected surplus per unit invested after wages, principal and benchmark interest, before the fixed premium and expansion costs.

Write k_F and $k_V(\tilde{\theta}^m)$ for optimal investment under the two contracts, suppressing the market index. With fixed finance, the interior choice is $k_F = (R - s^m + v_F)/\kappa$: a higher premium directly reduces project size. With variable finance, expanding the project also increases exposure to uncertain repayments, so optimal investment solves

$$\kappa k_V(\tilde{\theta}^m) + \sigma \tanh[\tilde{\theta}^m \sigma k_V(\tilde{\theta}^m)] = R.$$

The second term is the marginal risk cost of a larger floating-rate loan. Having optimised scale under each contract, borrowers compare the resulting certainty equivalents. The indifferent type $\theta^{m,*}$ separates those who float from the more risk-averse types who fix. Appendix E.1 derives the investment conditions and this contract comparison.

Aggregating these choices links loan demand to local investment. Let F_θ^m be the cumulative risk-aversion distribution and $f_F^m = 1 - F_\theta^m(\theta^{m,*})$ the fraction choosing fixed finance, with $\theta^{m,*} = \theta^*(s^m; R)$. Subscripts F, V identify contract types, while superscript m identifies the market. Fixed lending equals borrower mass times take-up times project size:

$$Q_F^m = \varphi^m f_F^m \frac{R - s^m + v_F}{\kappa}.$$

Variable loan sizes vary across borrower types t, so $Q_V^m = \varphi^m \int_0^{\theta^{m,*}} k_V(t)\,\mathrm{d}F_\theta^m(t)$. Together these loans finance local investment $I^m = Q_F^m + Q_V^m$.

The key real effect comes from both project scale and contract switching. At a given wage and benchmark rate, the indifferent borrower invests more under fixed finance: $k_F > k_V(\theta^{m,*})$. A higher premium therefore shrinks projects among borrowers who remain fixed and induces borrowers at the margin to switch to smaller floating-financed projects. Along

borrower demand, holding the wage and benchmark rate fixed,

$$-1 < \frac{\partial Q_V^m}{\partial Q_F^m} < 0, \qquad \frac{\partial I^m}{\partial s^m} < 0. \tag{19}$$

Floating credit replaces less than the lost fixed credit under these conditions (Appendix E.1). This demand-side derivative is not a general-equilibrium sign restriction: wages, household saving and bank competition also adjust. Since hours are proportional to project size at a given wage, the same pricing decisions affect labour demand.

4.2 Bank pricing and participation

Banks choose how much rate protection to supply to borrowers and how much to obtain through outside hedging. Consider bank i in market m. We focus on equilibria that are symmetric across banks within each market and therefore suppress the bank index i throughout, retaining superscript m to distinguish markets.

Household deposits $\mathcal{D}$ are allocated across markets in fixed shares δ^m. Local deposits are therefore $\mathcal{D}^m = \delta^m \mathcal{D}$, and each bank receives $d^m = \mathcal{D}^m / n^m$. The bank's fixed- and variable-rate loan volumes are q_F^m and q_V^m, and its net interbank lending is x^m. The latter is positive when the bank lends and negative when it borrows. Its balance sheet satisfies

$$q_F^m + q_V^m + x^m = d^m.$$

Variable loans and interbank positions earn $r(z)$; only variable loans generate net ancillary-service income b_V per unit.

In practice, banks have alternative hedging means. A bank may also buy a receive-fixed/pay-floating hedge of notional $\eta^m d^m \geq 0$, struck at r_0. Its payoff is $-\eta^m d^m \sigma z$ and its intermediation cost is $a\eta^m d^m$ terminal goods, with common $a > 0$. Competitive risk-neutral outside counterparties absorb the opposite payoff. The hedge has no initial funding requirement.

Borrowers can approach only banks in their lending segment, while all n^m banks compete for deposits across market m. This distinction is motivated by the role of branch networks and local competition in credit access (Aguirregabiria et al., 2025).[6] With J identical segments, each contains φ^m / J borrowers with the same type distribution and n^m / J banks. Segment inverse demand is $s^m(JQ_{F,\ell}^m; R)$, where $Q_{F,\ell}^m$ is segment fixed lending. At symmetry, its slope is Js_Q^m, where $s_Q^m = \partial s^m(Q_F^m; R)/\partial Q_F^m$. We extend these symmetric conditions continuously to $J \geq 1$, interpreted as effective segmentation: larger values mean fewer accessible

[6] Singh and Vives (1984) study price and quantity competition with differentiated products, while Weyl and Fabinger (2013) develop a general framework for imperfect competition and pass-through. Here the pricing multiplier arises from restricted access to otherwise identical lenders under Cournot competition within segments; borrower preferences over available contracts are unchanged.

lenders, n^m/J (Appendix E.1).

Taking funding, wages and rivals' fixed quantities within its segment as given, the bank chooses fixed lending and outside hedging before z is realised. These are individual choices; symmetry is imposed after deriving the bank's optimality conditions. Bank equity is $E^m = ed^m$, where $e > 0$ is a common equity-to-deposit scale used to normalise risk bearing, not an additional funding flow. With common CARA coefficient $\gamma > 0$ over return on equity, its problem is

$$\max_{q_F^m, \eta^m \geq 0} -\mathbb{E}\left[\exp\left(-\gamma \frac{\Pi^m(z)}{E^m}\right)\right], \tag{20}$$

where s^m denotes its segment premium (common across segments at symmetry) and operating profit before fixed overhead is

$$\Pi^m(z) = d^m(1-\beta^m)[r(z) - \underline{r}] + q_F^m[r_0 + s^m - r(z)] + b_V q_V^m - d^m \eta^m(\sigma z + a).$$

Changes in segment variable lending accrue to the bank changing fixed lending; interbank positions clear its balance sheet, with $q_V^m \geq 0$. Thus $\frac{\partial q_V^m}{\partial q_F^m} = \frac{\partial Q_V^m}{\partial Q_F^m}$, with the market-demand derivative given in (19).

Note that fixed lending and outside hedging both reduce the bank's exposure to rising rates, but expanding fixed lending also lowers the premium on its existing loans. At symmetry, residual earnings exposure per deposit is

$$G^m = 1 - \beta^m - \frac{Q_F^m}{\mathcal{D}^m} - \eta^m, \qquad \widetilde{\gamma} = \gamma/e.$$

Compared with the simple model's exposure in (5), the fixed-loan share is replaced by fixed credit per unit of deposits, $Q_F^m/\mathcal{D}^m$: project sizes are now endogenous, and interbank funding allows local lending to differ from local deposits. The additional term $-\eta^m$ captures outside hedging, so banks can reduce their earnings exposure without extending more fixed-rate credit to entrepreneurs. Differentiating the bank's certainty equivalent with respect to fixed lending, and using the envelope theorem for its hedge choice, gives the interior pricing condition (Appendix E.1):

$$s^m = -\frac{JQ_F^m}{n^m} s_Q^m - b_V \frac{\partial Q_V^m}{\partial Q_F^m} - \sigma \tanh(\widetilde{\gamma}\sigma G^m).$$

The first term is the markup from lending power: fewer accessible lenders make banks more reluctant to expand fixed credit and lower the premium on existing loans. The second compensates for foregone variable-loan ancillary income. The third lowers the premium when additional fixed lending hedges positive residual exposure, $G^m > 0$. External hedging can substitute for this motive without eliminating the markup. The effective lending-power index is J/n^m within a segment, whereas whole-market HHI is $1/n^m$.

Outside hedging limits this last incentive. Its optimality condition is

$$0 \leq \eta^m \;\perp\; a - \sigma \tanh(\widetilde{\gamma}\sigma G^m) \geq 0.$$

The bank buys a hedge when its marginal risk benefit would otherwise exceed the unit cost a. Therefore, with positive outside hedging, this benefit equals a, and loan pricing simplifies to

$$s^m = -\frac{JQ_F^m}{n^m} s_Q^m - b_V \frac{\partial Q_V^m}{\partial Q_F^m} - a, \qquad \eta^m > 0.$$

As a result, the bank can adjust its outside position to manage risk while continuing to restrict fixed credit to preserve lending rents. The hedge caps the marginal risk benefit of fixed lending, rather than its premium; borrower demand and loan market power still govern the price of protection.

These lending and hedging choices also determine the returns to entry. In each market, banks enter until the symmetric bank's profit certainty equivalent equals the local operating hurdle $F^m > 0$:

$$-\frac{E^m}{\gamma} \log \mathbb{E}\left[\exp\left(-\gamma \frac{\Pi^m(z)}{E^m}\right)\right] = F^m. \tag{21}$$

The hurdle includes fixed overhead in terminal goods and can differ across markets. Free entry makes net entry value zero in each active market; we impose no additional condition equating gross bank values across markets.

In the benchmark-rate experiment, a higher r_0 at fixed $\underline{r}$ initially widens the funding margin $(1-\beta^m)(r_0-\underline{r})$ and encourages entry. More lenders then reduce the markup, while the associated rise in deposit pass-through reduces banks' exposure to rate risk. When outside hedging remains active, banks can respond by reducing their hedge; once it reaches zero, the weaker risk benefit of fixed lending can offset the competitive pressure on premiums. Entry thus connects deposit competition to lending power and the supply of borrower protection.

4.3 Households

Household deposits fund lending, while labour supply determines the wage feedback to project profitability. These choices connect banking conditions to aggregate activity. A representative household chooses current consumption C_0, deposits $\mathcal{D}$, hours L_t at dates $t = 0, 1$, and terminal consumption $C_1(z)$. Deposits and hours are chosen at date 0, before the rate-shock state z is realised. Taking wages, deposit returns and retail dividends as given, it solves

$$\max_{C_0, \mathcal{D}, L_0, L_1, \{C_1(z)\}} \left\{ \log C_0 - \frac{\chi_0 L_0^{1+\nu}}{1+\nu} + \xi \left(\mathbb{E}[C_1(z)] - \frac{\chi_1 L_1^{1+\nu}}{1+\nu} \right) \right\}, \tag{22}$$

subject to the budgets below and nonnegative choices, with $C_0 > 0$. Here $\xi \in (0,1)$ is the discount factor, $\chi_t > 0$ scales labour disutility at date t, and the common $\nu > 0$ determines its curvature at both dates. Expectations average over the two rate states. Linear terminal utility makes depositors rank returns by expected repayment; household welfare therefore does not directly measure gains from reducing terminal payout risk.

Let w_t denote the common real wage and Ω_t^R retail dividends at date t. Fixed deposit relationship shares δ^m sum to one and need not equal borrower shares φ^m. The household's gross deposit return is $R_d(z) = \sum_m \delta^m [1 + r_d^m(z)]$. Its budgets are

$$C_0 + \mathcal{D} = w_0 L_0 + \Omega_0^R. \tag{23}$$

$$C_1(z) = w_1 L_1 + R_d(z)\mathcal{D} + \Omega_1^R, \qquad z \in \{-1, 1\}. \tag{24}$$

The household owns retailers; entrepreneurs and bank owners consume their own profits. At an interior optimum, an extra unit saved costs $1/C_0$ in current marginal utility and yields discounted expected repayment $\xi \mathbb{E}[R_d(z)]$, giving

$$\frac{1}{C_0} = \xi \, \mathbb{E}[R_d(z)]. \tag{25}$$

Higher deposit pass-through makes the expected return more responsive to the benchmark rate. A higher expected return therefore lowers current consumption at an interior optimum, releasing resources for saving at a given income. Equilibrium deposits also depend on wage income and dividends, however, and must equal the investment financed by banks.

Labour supply completes this connection to production. At each date, the household equates marginal disutility to wage income valued at marginal utility of consumption:

$$\chi_0 L_0^\nu = w_0 / C_0, \qquad \chi_1 L_1^\nu = w_1.$$

Lower current consumption raises the incentive to work at a given wage. At date 1, marginal utility of consumption is one, so labour supply depends only on the real wage and labour preferences. Together with entrepreneurs' labour demand, this condition determines how changes in investment translate into wages and employment. The terminal wage clears the market for committed hours at date 0; it adjusts across equilibria, before the rate shock is realised.

4.4 Aggregation and equilibrium

Credit and funding. Loan prices determine project sizes and hence local investment, $I^m = Q_F^m + Q_V^m$. Aggregating bank balance sheets connects these lending decisions to household saving. Since outside hedges require no initial funding, aggregate investment I equals

household deposits:

$$I = \sum_{m=1}^{M} I^m = \mathcal{D}. \tag{26}$$

Local lending need not equal local deposits: net interbank lending from market m is $X^m = n^m x^m = \mathcal{D}^m - I^m$. Market clearing requires $\sum_m X^m = 0$, so funding surpluses finance other markets' deficits while total credit remains tied to aggregate saving. The aggregate fixed share of loan volume is $L_F^{\text{agg}} = \sum_m Q_F^m / I$, while borrower take-up is $f_F^{\text{agg}} = \sum_m \varphi^m f_F^m$; these differ because project sizes depend on the financing contract and borrower type. Section 5 reports loan-volume shares. Fixed and floating credit may move in opposite directions, but their sum must equal endogenous deposits.

Production and labour. Competitive date-0 producers have productivity A_0, produce aggregate output $Y_0 = A_0 L_0$, and earn zero profits at intermediate-good price $p_0 = w_0/A_0$. At date 1, all projects face the same wage and therefore use the same hours per unit invested. Aggregate labour demand and output are

$$L_1 = \left[\frac{(1-\alpha)\bar{p}A_1}{w_1}\right]^{1/\alpha} I, \qquad Y_1 = A_1 I^{\alpha} L_1^{1-\alpha}. \tag{27}$$

At a given wage, lower investment therefore reduces terminal labour demand. Household labour supply then determines the adjustment of the common wage and employment, feeding back into project profitability. This wage feedback transmits lending decisions through the real economy; it is not an additional source of banking market power. These quantities are determined at date 0, when hours are committed.

Monopolistically competitive retailers convert intermediate goods one for one into final consumption and investment goods. With the final-good price normalised to one, flexible terminal pricing gives $p_1 = \bar{p} = (\varepsilon - 1)/\varepsilon$. Preset current retail prices allow p_0 to adjust after a rate change. Retail dividends at date $t \in \{0, 1\}$ are

$$\Omega_t^R = (1 - p_t) Y_t. \tag{28}$$

Retail dividends enter household income, completing the link from production to consumption and saving. Appendix E.2 derives pricing and the baseline condition $p_0 = \bar{p}$.

Goods markets. Current goods clearing requires production to meet consumption and investment demand:

$$Y_0 = C_0 + I. \tag{29}$$

Together with $Y_0 = A_0 L_0$, this condition connects changes in household consumption and bank-financed investment to current employment. Let $C_1^E(z)$ and $C_1^B(z)$ denote terminal

consumption of entrepreneurs and bank owners; $C_1(z)$ is household consumption. Write $J_2 = \sum_m \varphi^m \int_0^\infty k^m(\theta)^2 \, \mathrm{d}F_\theta^m(\theta)$ for the aggregate second moment of project size, where $k^m(\theta)$ is investment chosen by type θ under its preferred contract. In each state,

$$Y_1 + b_V \sum_m Q_V^m = C_1(z) + C_1^E(z) + C_1^B(z) + \frac{\kappa}{2} J_2 + \sum_m n^m F^m + \mathcal{H} + \Lambda(z), \quad z \in \{-1, 1\}. \tag{30}$$

Here $\mathcal{H} = a \sum_m \mathcal{D}^m \eta^m$ is the resource cost of hedging and $\Lambda(z) = \sigma z \sum_m \mathcal{D}^m \eta^m$ is the net payment to outside counterparties, with zero expectation. Net ancillary-service output $b_V \sum_m Q_V^m$ adds to project output; expansion costs $\kappa J_2/2$ and bank overhead $\sum_m n^m F^m$ use resources. Note that loan and deposit repayments redistribute income within the economy, whereas entry, project expansion and outside hedging affect the resources available for consumption. The nonpecuniary benefit $v_F k$ is excluded from goods accounts. Appendix E.3 verifies this identity state by state.

Definition 1 (General equilibrium conditional on the benchmark and outside rates)**.** *Given the rate process, outside return, borrower populations, deposit shares, segmentation J, search frictions, entry hurdles and inherited retail contracts, an interior equilibrium consists of local bank counts, prices, wages, consumption, employment, investment and financial portfolios such that:*

1. *The household maximises* (22) *subject to* (23) *and* (24)*, taking prices, dividends and deposit shares as given.*

2. *Entrepreneurs choose contract, scale and committed employment. Producers, retailers and the final-good aggregator optimise at inherited current retail prices and flexible terminal prices.*

3. *In each market $n^m > 1$, with $n^m/J \geq 1$, satisfies* (21) *at the anticipated lending equilibrium. Given n^m, deposit rates follow* (16) *and lending and hedging maximise bank value and satisfy the restricted-access pricing and hedge conditions, with $q_F^m = Q_F^m/n^m$ and $q_V^m = Q_V^m/n^m$.*

4. *Bank balance sheets and loan markets clear, interbank positions sum to zero, and* (26)*,* (29)*,* (27) *and* (30) *clear funding, goods and labour markets. Consumption and repayments are feasible in both states.*

Under symmetry each lender originates a share $1/n^m$ of market-wide credit, so the measured lending Herfindahl index is $1/n^m$. Concentration among the lenders accessible within a segment is instead J/n^m.

The calibration first quantifies entry, loan prices, contract composition and investment after a benchmark-rate increase. We then vary risk tolerance and hedging conditions to isolate banks' incentives to supply borrower protection.

5 Quantitative results

How does bank market power translate changes in interest rates and risk into investment, employment and household welfare? Three experiments address this question from a common two-market calibration. A higher benchmark rate changes financing costs and competition through entry; greater uncertainty changes the demand for borrower protection and bank hedging; and greater bank risk-bearing capacity isolates banks' willingness to supply that protection at fixed market structure. In each case, loan pricing affects project scale, while contract substitution, wages and household saving determine the aggregate response.

5.1 Specification and calibration

We use two markets with different funding sensitivities and lending concentration to allow competition and hedging incentives to produce different contract responses. Deposit betas are 0.26 and 0.44, and each market holds half of household deposits. Earlier three-market exercises yield the same qualitative monetary and risk-bearing-capacity results. Entrepreneur populations are fixed across counterfactuals, with their relative masses estimated jointly with financial preferences. Baseline HHIs determine effective bank counts and deposit betas determine search frictions:

$$n^m = 1/H^m, \qquad \zeta^m = \frac{\beta^m}{(1-\beta^m)(n^m-1)}.$$

HHI refers to the whole market; borrowers access n^m/J lenders. Entry hurdles follow from baseline free entry and remain fixed thereafter.

Borrower risk aversion is lognormal in each market. We report the market medians e^{μ^L} and e^{μ^H} alongside the log dispersions τ^L and τ^H, imposing equal arithmetic means:

$$e^{\mu^L+(\tau^L)^2/2} = e^{\mu^H+(\tau^H)^2/2} = \bar{\theta}.$$

The medians can therefore differ even though average risk aversion is the same. These four distribution parameters provide three free parameters; together with γ, J, b_V, v_F, a and φ^L, there are nine parameters to discipline jointly.

Baseline premiums and fixed shares discipline the balance between borrowers' demand for protection and banks' willingness to supply it. Responses to a 50 bp rate rise reveal how entry and deposit competition alter that balance. The provisional baseline hedge-to-deposit ratio of 1.47% and the aggregate fixed-share response of -0.66 pp to a 25 bp uncertainty increase discipline substitution between fixed lending and outside hedging. The latter comes from the bank-fixed-effects estimate controlling for interest rates. Because borrower preferences, lending power, bank risk aversion and hedge costs jointly affect prices and quantities, we fit

the nine parameters together to these nine moments (Table 2).[7]

Table 2: Parameters and calibration targets

Parameter	Value	Target or calibration basis
Jointly fitted financial parameters		
Borrower medians (e^{μ^L}, e^{μ^H})	(19.32, 0.80)	*Nine joint moments:*
Log dispersions (τ^L, τ^H)	(2.23, 3.37)	Baseline premiums (bp): $(22.20, 6.10)$;
Bank risk aversion γ	7.89	fixed shares (%): $(17.20, 21.20)$.
Lending segmentation J	3.94	After a 50 bp rate rise: aggregate
Bank floating benefit b_V (bp)	11.94	premium -3.00 bp, fixed share $+1.30$
Borrower benefit v_F (bp)	4.63	pp, HHI -0.067.
Outside hedge cost a (bp)	9.48	Baseline outside hedges/deposits: 1.47%.
Population share φ^L (%)	65.31	After a 25 bp uncertainty rise: aggregate fixed share -0.66 pp.
Search and entry parameters		
Deposit search ζ^m	(0.11, 0.15)	Baseline betas $(0.26, 0.44)$, given baseline HHIs
Entry hurdles F^m (bp)	(7.07, 3.11)	Free entry at HHIs $(0.24, 0.16)$
Maintained assumptions		
Equity/deposit ratio e	0.10	Risk-bearing normalisation
Rate uncertainty σ (bp)	50.00	Maintained rate uncertainty
Mean real rate r_0 (%)	-0.75	Nominal benchmark less inflation
Real outside return $\underline{r}$ (%)	-1.34	Implied by the 59 bp rate gap
Rate gap $r_0 - \underline{r}$ (bp)	59.00	Imposed funding–saving gap
Deposit shares δ^m	1/2	Equal deposit relationship shares
Capital share α	1/3	Maintained production curvature
Expansion curvature κ	0.10	Maintained investment curvature
Retail elasticity ε	6.00	Maintained retail markup
Discount factor ξ	0.96	Maintained household preference
Labour exponent ν	2.00	Common labour-supply curvature at both dates
Baseline normalisations		
Project productivity A_1	3.94	Baseline $I^b = L_1^b = 1$
Current productivity A_0	2.05	Baseline $L_0^b = 1$ and goods clearing
Labour scale χ_0	1.62	Baseline $p_0^b = (\varepsilon - 1)/\varepsilon$
Labour scale χ_1	2.19	Baseline $L_1^b = 1$ and labour clearing
Baseline surplus R^b	0.10	Baseline $I^b = 1$ with joint fit

Notes: Market values are ordered low/high beta. The two medians and two dispersions satisfy the equal-mean restriction, with $\bar{\theta} = 230.46$, leaving three free distribution parameters and nine jointly fitted parameters in total; $\varphi^H = 1 - \varphi^L$. Entry hurdles are per-bank resource costs reported as basis points of baseline aggregate investment, $10^4 F^m / I^b$, with $I^b = 1$. The model HHI response is -0.06523, compared with the empirical target of -0.067; the other eight targets are matched to numerical precision. Premium and HHI responses use baseline loan-volume weights. The hedge target measures net receive-fixed exposure relative to deposits, not gross derivative notional, and is provisional. Values are rounded for presentation.

[7]The choice of calibration also takes account of the market uncertainty responses, so these are not fully independent validation moments. The selected fit is close to a bank-choice indifference boundary.

Baseline investment, deposits and hours at both dates are normalised to one. Given the jointly fitted project surplus, the productivity scales A_t and labour-disutility scales χ_t support this allocation through goods and labour clearing (Appendix E.2). Their date-specific values accommodate the different production and consumption roles of the two periods. They remain fixed across counterfactuals, allowing us to focus on how banking conditions affect real activity rather than fitting additional real-side levels. Labour curvature is common across dates, $\nu = 2$. Production and household curvatures, the mean real benchmark rate of -0.75% and the 59 bp benchmark–outside-return gap retain their earlier values. Segmentation uses the continuous approximation ($J = 3.94$): baseline n^m/J is approximately 1.06 and 1.59. These are effective competition indices, not directly observed counts of accessible banks. Hedge costs are linear.

Baseline outside hedging is 2.94% of low-beta deposits and zero in the high-beta market. The uncertainty experiment selects positive hedging in both markets, subject to bank-choice checks. Market-specific monetary and uncertainty responses provide comparisons rather than exact targets. Investment, employment, welfare and the risk-tolerance experiment are implications of the fitted model. The aggregate monetary HHI decline is approximated (-0.06523 against -0.067); the other eight moments are matched to numerical precision.

5.2 Bank competition and monetary transmission

We interpret this experiment as the medium-run response to a higher mean benchmark rate r_0, allowing bank entry, loan pricing and borrowers' financing choices to adjust. Deposit power initially widens funding margins by $(1 - \beta^m)\Delta r_0$ when the outside return is fixed, attracting entry and reducing concentration. Entry then weakens lending power and compresses markups, but also raises deposit pass-through, reducing banks' exposure to rate risk and their incentive to supply fixed loans as a hedge. These opposing effects determine how much cheaper protection cushions investment against higher borrowing costs. Figure 5 compares the adjusted equilibria for rate increases from 0 to 100 bp; its diamonds mark the three aggregate monetary targets at +50 bp.

Panels A–C show how entry translates into different pricing responses across markets. At +50 bp, competition dominates in the aggregate: the premium falls by 3.00 bp and the fixed share rises by 1.30 pp. The low-beta market experiences a larger initial funding-margin gain and a greater decline in concentration; its premium falls by 5.65 bp, raising the fixed share by 5.53 pp. In the high-beta market, the weaker hedging incentive outweighs the pressure on markups: the premium rises by 1.95 bp and the fixed share falls by 6.61 pp. Lower concentration therefore need not make protection cheaper in every market.

These market-specific responses agree with the data in direction and ordering, although the model generates larger fixed-share movements and smaller HHI declines (Table 4). Hold-

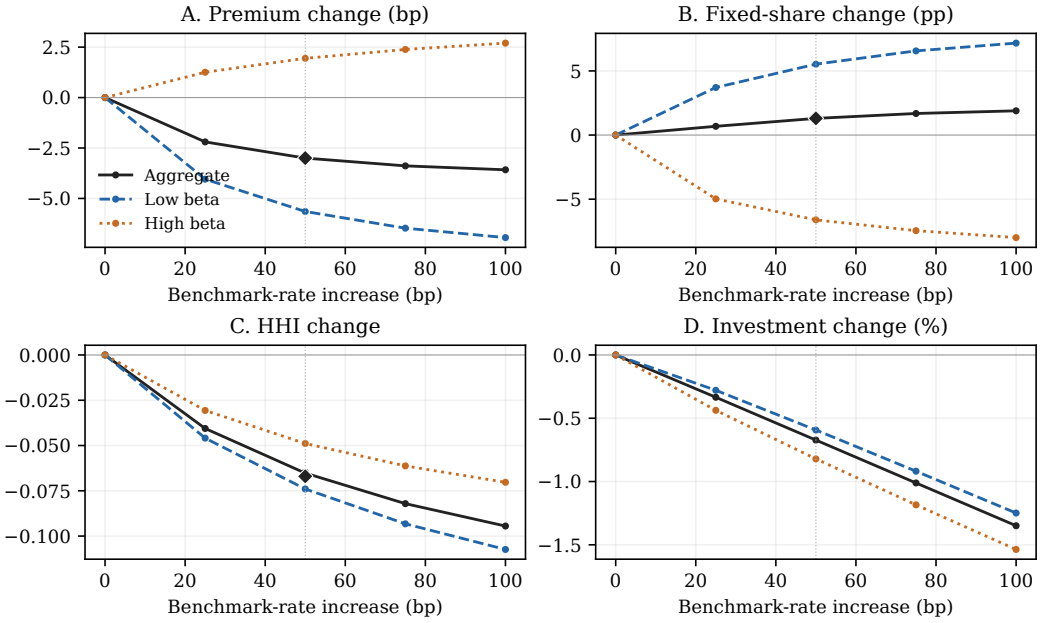


Figure 5: Responses to benchmark-rate increases

Notes: Changes are relative to baseline. Premiums and HHIs use baseline loan-volume weights; aggregate fixed shares use current volumes. Diamonds mark aggregate premium, fixed-share and HHI targets. Entry adjusts at every point.

ing bank counts fixed produces almost no aggregate repricing or contract substitution: the premium falls by 0.05 bp and the fixed share changes by less than 0.01 pp. Entry, and the associated change in lending power, therefore accounts for most of the financial adjustment. This mechanism depends on the relative rate change: allowing the outside return to rise with the benchmark attenuates entry and contract substitution (Appendix E.4).

Table 3 separates these effects on aggregate investment. Allowing entry while holding deposit betas fixed lowers the premium by 6.24 bp and raises the fixed share by 9.19 pp. Restoring the beta response weakens the earnings-hedge motive and reduces the funding gains supporting entry, leaving the targeted 3.00 bp premium decline and 1.30 pp share increase. Investment falls by 0.686% with fixed counts, 0.656% with entry and fixed betas, and 0.673% in the full equilibrium. Thus entry initially cushions the contraction by 0.030 pp, while the beta response offsets 0.016 pp: the net cushioning is only 0.014 pp. In this medium-run comparison, competition changes the composition of credit much more than its total volume.

The aggregate contraction reflects higher financing costs under both contracts. The 3.00 bp premium decline offsets only a small part of the benchmark increase: the fixed rate rises by 47.00 bp on average and the expected floating rate by 50 bp. Entrepreneurs consequently reduce project sizes in both markets, explaining the investment contraction in Panel D despite their opposite fixed-share responses. Lower investment reduces labour demand and income; wage adjustment cushions project returns, while household saving falls with the funding

Table 3: Monetary transmission through entry and deposit competition

	Δs (bp)	ΔL_F (pp)	ΔQ_F (%)	ΔI (%)	ΔY_1 (%)	ΔHHI
Fixed counts	-0.05	0.00	-0.69	-0.686	-0.295	0.0000
Entry, fixed betas	-6.24	9.19	48.43	-0.656	-0.282	-0.0821
Full response	-3.00	1.30	6.27	-0.673	-0.289	-0.0652

Notes: All rows raise r_0 by 50 bp with the outside return fixed. Fixed counts also fix deposit betas. The second row allows entry but holds betas at baseline, overriding the search rule as a diagnostic intervention. The last row restores that rule. Premium and HHI changes use baseline loan weights; shares use current volumes. Y_1 is terminal output. Differences are evaluated in the displayed order, not an order-independent decomposition. All rows pass numerical bank-choice checks.

Table 4: Heterogeneous responses to a 50 bp rate increase

	Low beta		High beta	
	Model	Data	Model	Data
Fixed-share change (pp)	5.53	3.10	-6.61	-5.70
HHI change	-0.074	-0.14	-0.049	-0.09
Total investment change (%)	-0.59	—	-0.82	—
Average entrepreneur CE change (%)	-1.32	—	-1.61	—

Notes: Market-specific monetary responses are not targeted; the calibration uses aggregate premium, fixed-share and HHI responses. Model premiums change by -5.65 bp in the low-beta market and $+1.95$ bp in the high-beta market. Shares refer to loan volumes. Dashes indicate outcomes without empirical counterparts reported here. Entrepreneur populations remain fixed.

needed by firms. Aggregate investment and deposits decline by 0.67%, current consumption by 0.29% and terminal employment by 0.10% (Table 5). Fixed lending nevertheless rises by 6.27%, while floating lending falls by 2.26%. The expansion of protected credit therefore accompanies a contraction in aggregate activity.

These higher financing costs reduce entrepreneurs' optimised profit certainty equivalents, including the nonpecuniary contract benefit, by 1.32% in the low-beta market and 1.61% in the high-beta market, giving a population-weighted decline of 1.42%. Cheaper protection cushions the low-beta loss. Household utility falls by the equivalent of 0.47% of baseline current consumption, holding other baseline allocations fixed: the consumption loss outweighs the benefit of fewer hours. Entrepreneur CE and household consumption equivalents are distinct welfare measures, reported separately; Section 5.3 defines a common monetary measure for the cheaper-hedging comparison.

5.3 Uncertainty, risk sharing and access to lenders

Higher expected rates reduce investment by raising financing costs. Greater uncertainty can also discourage project expansion, but its effect depends on how banks supply protection. Holding expected rates and other primitives fixed, a higher σ strengthens both borrowers'

demand for protection and banks' hedging needs. The former pushes premiums up; the latter can push them down through demand for fixed loans. Outside hedging caps banks' marginal hedging benefit at a, allowing them to stabilise earnings while preserving lending rents. The choice between fixed lending and external hedges therefore determines how much rate risk remains with investing firms.

Increasing σ from 50 to 75 bp raises the aggregate premium by 10.26 bp and reduces the fixed share by the targeted 0.66 pp (Table 6). Both markets then hedge externally, with notionals around 25% of deposits and positive residual earnings exposure. The marginal hedging benefit remains 9.48 bp in the low-beta market and rises from 8.05 to 9.48 bp in the high-beta market; it does not fall. Premiums rise by 11.90 and 7.21 bp, respectively, but fixed shares move in opposite directions. These market responses are not imposed as targets and agree in sign with the data, although the high-beta decline is larger.

The contrast reflects borrower heterogeneity as well as bank portfolio choices. Despite equal mean risk aversion, the high-beta market has greater dispersion and a lower median, leaving many borrowers relatively willing to retain floating-rate risk. In the low-beta market, outside hedging expands alongside fixed lending; higher premium income outweighs the

Table 5: Aggregate quantity responses to a 50 bp rate increase

Outcome	Change (%)
Investment and deposits	-0.67
Fixed lending	+6.27
Floating lending	-2.26
Current consumption	-0.29
Terminal employment	-0.10

Notes: Entry adjusts fully. Real outcomes depend on the maintained production and household preferences; contract volumes also reflect the fitted aggregate share response.

Table 6: Responses to an uncertainty increase from 50 to 75 bp

	Aggregate		Low beta		High beta	
Change	Model	Data	Model	Data	Model	Data
Fixed share (pp)	-0.66	-0.66	3.00	4.07	-7.47	-5.12
HHI	-0.0046	0.0064	-0.0082	-0.0146	0.0022	0.0080
Investment (%)	-0.07	—	-0.25	—	0.26	—
Entrepreneur CE (%)	0.16	—	-0.10	—	0.63	—
Household equivalent (%)	-0.08	—	—	—	—	—

Notes: Changes compare baseline with the selected equilibrium in which both markets hedge externally. The results characterise this selected hedging regime, not a unique response to uncertainty; a continuous adjustment path is not required for this static equilibrium comparison. Only the aggregate fixed-share response is targeted; market fixed-share and all HHI responses are comparison moments. Model aggregate HHI changes use baseline loan-volume weights; entrepreneur CE uses population weights. Dashes indicate empirical counterparts not reported here or inapplicable market entries. Household welfare is the representative household's current-consumption equivalent, not separate market welfare.

hedging burden and supports entry. In the high-beta market, banks expand outside hedges while reducing fixed lending; the smaller fixed-loan book limits income gains relative to hedging expenditure, inducing exit. Entry therefore restrains low-beta markups, while exit reinforces the high-beta contraction that dominates the aggregate fixed-share response.

These entry responses reproduce the market-level HHI directions, though their magnitudes are smaller than in the data. The low-beta decline dominates model aggregation, yielding -0.0046 against an empirical increase of 0.0064. Both changes are small in levels relative to the monetary experiment, so the main agreement is the cross-market pattern; the aggregate sign remains a discrepancy.

With the expected benchmark rate unchanged, most adjustment concerns contract composition: aggregate investment falls by only 0.07%. Lower wages raise project surplus, especially benefiting relatively risk-tolerant high-beta borrowers. Investment rises in that market despite its shift towards floating finance, and its CE gain outweighs the low-beta loss, raising average entrepreneur CE by 0.16%. Household welfare instead falls by the equivalent of 0.08% of baseline current consumption. With linear terminal utility, this loss operates through expected consumption and labour rather than a direct variance penalty.

The entrepreneur gain need not imply more efficient risk sharing. Following Section 3.5, we hold project sizes, wages, deposits, bank counts and deposit contracts fixed, then reassign loan contracts and optimise outside hedges to maximise combined borrower and bank monetary CE. At 75 bp, this conditional benchmark raises the fixed share to 53.73% and yields a gain of 3.42 bp per unit of investment, compared with 1.35 bp at baseline. The fixed-share shortfall is large, but the monetary gap is modest. This benchmark measures conditional risk sharing, not a full social-planner allocation or total general-equilibrium welfare. Greater uncertainty widens the gap despite the average entrepreneur CE gain.

In a separate comparison, improving lender access narrows this gap by strengthening loan competition. At 75 bp, reducing J by 25% lowers the premium by 10.79 bp, expands fixed lending and reduces the remaining conditional gap to 1.50 bp. The associated income effects redistribute the gains: household welfare rises by 0.05% while average entrepreneur CE falls by 0.17%, so better risk sharing alone does not establish an overall welfare gain. The checked lower-J equilibrium also has no high-beta outside hedge; this comparison does not demonstrate a transition from the selected equilibrium (Appendix E.5).

Cheaper outside hedging affects borrower protection differently from greater lender access. Returning to baseline uncertainty, we reduce a by 10%, keeping the calibration fixed and allowing entry to adjust. Banks substitute towards external hedges, weakening their incentive to supply fixed loans: the aggregate premium rises by 0.87 bp and the fixed share falls by 0.67 pp. Entry only partly offsets this effect, and investment declines slightly. Lower wages benefit high-beta entrepreneurs, whose loan premiums barely change, but low-beta entrepreneurs lose slightly as financing costs rise. Greater hedge use also raises total hedging

expenditure despite its lower unit cost. With equal weights on terminal-good equivalents, the household loss of 0.36 bp of baseline investment outweighs the average entrepreneur gain, worth 0.07 bp; net bank CE remains zero under free entry, leaving an aggregate welfare loss of 0.29 bp.[8] Thus the welfare consequences of improving banks' hedging options depend on how loan market power shapes the protection supplied to borrowers.

5.4 Bank risk tolerance and borrower protection

The preceding experiments link rate changes to real activity through financing costs and borrower protection. We now ask whether greater bank risk-bearing capacity supports investment when lending power governs the supply of that protection. Starting from the same baseline, we hold expected rates, uncertainty and bank counts fixed. Capacity c rises from 1 to 1.8, replacing γ by γ/c in the bank objective:

$$-\mathbb{E}\exp\left(-\frac{\gamma}{c}\frac{\Pi^m(z)}{E^m}\right).$$

This lowers effective risk aversion without injecting equity: $e = 0.10$ remains fixed. Deposit betas and accessible lender counts are unchanged, and free entry is not reimposed; portfolios and real allocations continue to adjust.

Fixed lending still offsets the mismatch between floating asset income and less responsive deposit costs, but greater risk tolerance reduces the value of this protection. Where outside hedging is active, banks initially reduce that position while the marginal hedging benefit of fixed lending remains a. Once the position reaches zero, further risk tolerance can lower this benefit and raise the required premium. Accordingly, the high-beta market, which starts without an outside hedge, responds more sharply in Panels B and C of Figure 6. At $c = 1.8$, neither market hedges outside; the aggregate premium rises by 7.58 bp and the fixed share falls by 7.13 pp. The high-beta premium rises by 13.74 bp and its share falls by 14.82 pp, compared with 4.28 bp and 3.02 pp in the low-beta market.

Table 7 shows why greater capacity does not necessarily stimulate lending. Expected pre-overhead earnings rise in both markets: in the aggregate, from 50.87 to 51.81 bp of deposits, while their standard deviation rises from 22.47 to 26.77 bp. Banks use the additional capacity to retain more earnings risk and supply less fixed-rate protection.

[8] We sum the household utility change divided by ξ, population-weighted entrepreneur CE changes and bank CE changes net of entry costs. Outside counterparties are competitive and risk neutral, with zero expected net surplus; the reduction in a lowers real unit costs rather than subsidising hedging. The checked equilibrium path connects to baseline.

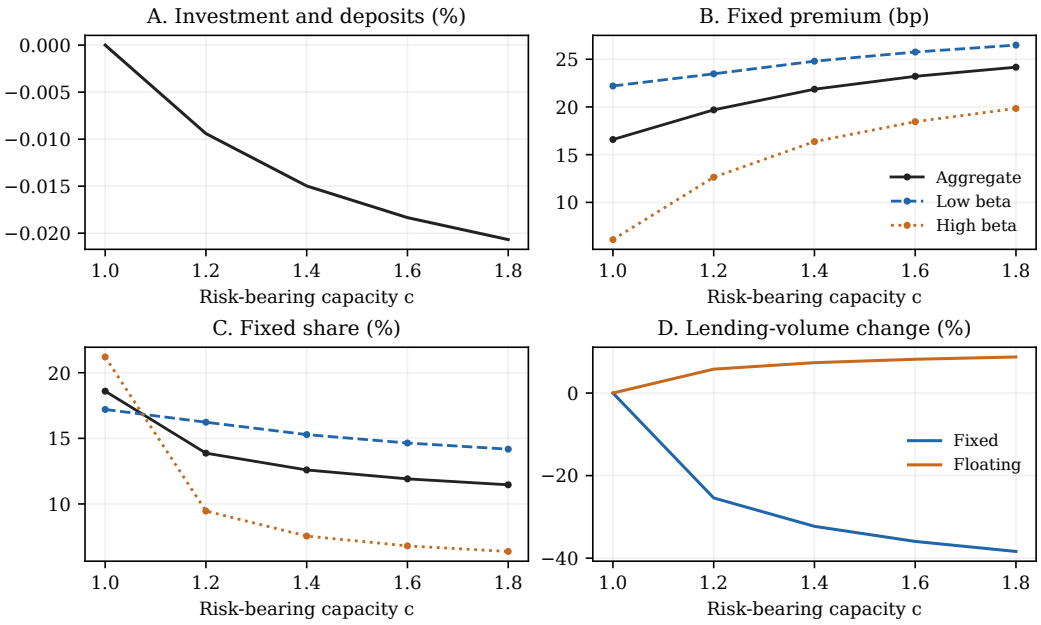


Figure 6: Bank risk-bearing capacity and lending

Notes: Capacity divides bank risk aversion by c. Counts, expected rates and all other primitives remain fixed. Premiums use baseline loan-volume weights; fixed shares use current volumes. Quantity changes are relative to $c = 1$.

Table 7: Bank earnings and risk as risk tolerance rises

c	Market	Earnings (bp)	SD (bp)	G	η (%)	s (bp)	ΔL_F (pp)
1	Low beta	61.24	24.33	0.487	2.94	22.20	0.00
1	High beta	40.50	20.61	0.412	0.00	6.10	0.00
1	Aggregate	50.87	22.47	0.449	1.47	16.59	0.00
1.4	Low beta	61.78	27.04	0.541	0.00	24.79	-1.91
1.4	High beta	41.59	25.37	0.507	0.00	16.37	-13.65
1.4	Aggregate	51.69	26.20	0.524	0.00	21.86	-6.00
1.8	Low beta	61.91	27.76	0.555	0.00	26.48	-3.02
1.8	High beta	41.71	25.78	0.516	0.00	19.84	-14.82
1.8	Aggregate	51.81	26.77	0.535	0.00	24.17	-7.13

Notes: Counts and expected rates are fixed. Earnings are $\mathbb{E}[\Pi^m]/d^m$, before overhead and including ancillary income and hedge costs; SD is $\sigma|G^m|$, both in basis points of deposits. Aggregate earnings and exposure use deposit weights; aggregate SD is $\sigma|\sum_m \delta^m G^m|$ because shocks are common. Premiums use baseline loan weights and shares use current volumes. Investment changes at $c = 1, 1.4, 1.8$ are $0, -0.015, -0.021\%$. Bank choice is checked globally on the numerical search domain. Changed risk preferences preclude interpreting earnings gains alone as welfare gains.

The resulting adjustment is mainly between contracts (Panel D): fixed lending falls by 38.38%, while floating lending rises by 8.74%. Higher premiums reduce project sizes among borrowers who continue to fix, and switching to floating finance exposes borrowers to risk that discourages investment. Floating credit therefore replaces slightly less than the lost fixed credit, leaving aggregate investment and deposits 0.02% lower (Panel A). This contraction is much smaller than in Section 5.2, because expected floating borrowing costs do not rise.

Unlike the monetary experiment, wage adjustment now raises average entrepreneur CE: the low-beta gain of 0.12% outweighs the high-beta loss of 0.03%, where protection becomes substantially more expensive, yielding a population-weighted gain of 0.07%. Household welfare nevertheless falls by the equivalent of 0.02% of baseline current consumption. Current consumption is unchanged, but lower investment reduces terminal income, outweighing the benefit of fewer hours.

Greater capacity, like cheaper outside hedging, need not stimulate investment or improve household welfare when banks use lending power to charge more for protection. Diversification or anticipated public support can motivate greater risk tolerance (Chari & Kehoe, 2016; Froot & Stein, 1998); we isolate this channel without modelling either intervention. Contract substitution and wage adjustment limit the aggregate effects in these exercises. We next examine monetary transmission when financing types and loan terms are inherited.

6 Bank power and short-run monetary transmission

In the medium-run monetary experiment, both fixed and variable borrowing rates adjust to the higher benchmark, so a larger fixed share provides limited insulation from the increase in financing costs. In the short run, inherited fixed coupons protect investment from that increase. We therefore examine how bank market power before a policy surprise shapes the subsequent contraction in investment, output and employment. The comparison changes the timing of financial adjustment within the same two-date economy, keeping the calibration and flexible wages unchanged.

For each value of J, we first solve the original equilibrium with mean benchmark rate r_0, keeping all other parameters unchanged. Within $t = 0$, an unanticipated increase $\Delta r = 50$ bp in the mean benchmark rate occurs after bank entry and loan contracting, but before drawdowns, investment and hiring. Bank counts and deposit pass-through coefficients are inherited, and borrowers retain their financing types and agreed rate schedules. They can increase or decrease drawdowns without borrowing ceilings. Fixed coupons remain unchanged, while variable rates reset:

$$r_F^m = r_0 + s_{\text{old}}^m, \qquad r_V(z) = r_0 + \Delta r + \sigma z.$$

Here s_{old}^m is the pre-shock premium; future uncertainty σ is unchanged. Wages, hours, investment and deposits adjust after the surprise but before z is realised at $t = 1$, using the household conditions and market clearing in Section 4. Deposit rates retain their inherited indexing rule, and interbank positions accommodate local funding imbalances.[9]

[9]Bank equity and outside hedge notionals and strikes are also inherited. Each unit of the existing receive-fixed/pay-floating hedge pays $-\Delta r - \sigma z$, with its original intermediation cost. Pre-shock bank choices pass numerical global lending and hedging optimality checks. The surprise is unanticipated when those choices

Table 8 compares the calibrated $J = 3.94$, a 25% reduction and $J = 1$. These are economies with different lender access before the same policy surprise. Lower J weakens banks' ability to restrict fixed lending for rents, reducing the price of protection and raising the initial aggregate fixed share from 18.59% to 24.62% and 35.02%, respectively. The resulting investment contractions are 0.56%, 0.52% and 0.45%: moving from calibrated J to $J = 1$ reduces the contraction by 0.112 percentage points, approximately 20%.

Table 8: A 50 bp surprise with inherited loan contracts

Response (%)	Calibrated J	$0.75J$	$J = 1$
Investment and deposits	−0.56	−0.52	−0.45
Consumption, $t = 0$	−0.18	−0.18	−0.18
Output and employment, $t = 0$	−0.36	−0.34	−0.31
Output, $t = 1$	−0.24	−0.22	−0.19
Employment, $t = 1$	−0.08	−0.07	−0.06
Fixed-financed investment	3.52	3.25	2.80
Variable-financed investment	−1.49	−1.75	−2.20

Notes: Percentage changes relative to each economy's own pre-shock equilibrium. Only J varies across columns; calibrated $J = 3.9414$. Entry and loan terms adjust to J before the surprise and remain fixed thereafter. Both wages and deposits adjust; financing types are inherited and drawdowns have no ceiling.

Inherited protection operates through both financing costs and wage feedback. Variable borrowers face higher expected repayments and reduce investment, lowering labour demand. Falling wages support project returns for both types, but fixed borrowers retain their old coupon and can therefore expand investment as labour becomes cheaper. Lower J leaves more borrowers in this protected group, strengthening its contribution to aggregate investment. As a result, the economy contracts less and wages fall less; the remaining variable borrowers receive less cost relief and contract more at $J = 1$, despite the milder aggregate decline.

This investment channel carries through to aggregate activity. At $t = 0$, lower investment reduces demand for goods and labour; at $t = 1$, the smaller capital stock reduces production and employment. Household income and deposits fall alongside investment. Because inherited deposit pass-through differs little across J, the consumption response at $t = 0$ is almost unchanged, so stronger lending competition cushions output mainly through investment. Bank market power thus affects monetary transmission by determining how much spending is protected before rates rise. This short-run effect survives wage adjustment and includes fixed borrowers' ability to expand drawdowns at inherited coupons; the original retail-pricing arrangements remain in place.

are made; we do not reimpose post-shock pricing or free entry.

7 Conclusion

This paper shows how relative market power in funding and lending shapes banks' interest rate risk and monetary transmission. Banks with less rate-sensitive funding charge higher fixed–variable loan premiums, even though stable funding makes fixed-rate lending an attractive earnings hedge. Greater lending concentration is associated with higher premiums conditional on funding sensitivity and a stronger expansion of fixed-rate loan stocks after monetary tightening. Our model explains these patterns through banks' tradeoff between earning lending rents and hedging earnings risk. Entry links competition on both sides of the balance sheet, changing both the incentive to hedge and the price of borrower protection.

The joint calibration connects this mechanism to household saving and investment. Following a higher real financing rate, entry lowers aggregate concentration and the fixed-rate premium, while the fixed share of lending rises. Fixed credit expands even as total investment and household deposits contract: cheaper protection partly cushions, but does not offset, higher overall financing costs. The responses differ across markets because stronger lending competition and weaker funding hedges have opposing effects on fixed-rate pricing.

Using the same calibration, greater bank risk-bearing capacity raises premiums and reduces fixed-rate lending when expected rates and bank counts are held fixed. Banks become less willing to supply protection as a hedge, and borrowers switch towards floating finance and reduce project scale. Investment falls as a result. Diversification and anticipated public support may operate through this channel, making lending terms and borrower investment relevant to assessing changes in bank resilience.

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A Data Appendix

A.1 U.S. Small Business Administration 7(a) loans

We use loan-level administrative records from the SBA's 7(a) programme, which supports small businesses unable to obtain conventional credit on reasonable terms. Participating lenders originate loans, while the SBA guarantees a portion of each loan. Proceeds finance working capital, equipment, business premises and acquisitions, as well as refinancing existing debt. Loans may carry fixed or variable interest rates, negotiated between lenders and borrowers subject to programme limits. For variable-rate loans, the repricing frequency is also negotiated between borrower and lender, subject to the requirement that rates adjust no more frequently than monthly. For a detailed overview of the programme, its borrowers and lenders, and loan characteristics, see Gong et al. (2025).

We focus on loans approved during 2010Q1–2019Q4 under the main *Guaranty* subprogramme and issued by FDIC-insured depository institutions. The records identify the lender and report the initial interest rate, rate type, approved and guaranteed amounts, maturity, and borrower location, industry and jobs supported. Lender identifiers allow us to connect loan terms to banks' regulatory reports and branch networks. About one fifth of loans carry fixed rates, both by number and value. Table 9 provides detailed summary statistics.

Table 9: Descriptive statistics of SBA 7(a) loans by rate type, 2010Q1–2019Q4

	Fixed-rate				Variable-rate			
	Mean	P25	P50	P75	Mean	P25	P50	P75
Number of banks	1,013				1,643			
Number of loans	33,207				115,741			
Gross approval ($ billions)	27.8				97.2			
Approval amount ($ thousands)	837	294	560	1,071	840	239	492	1,053
Guarantee share (%)	76.7	75.0	75.0	75.0	77.6	75.0	75.0	75.0
Interest rate at origination (%)	5.86	5.25	5.85	6.50	6.08	5.50	6.00	6.65
Maturity (months)	182	116	120	300	172	120	120	300
Jobs supported	15.8	3.0	7.0	18.0	15.5	3.0	8.0	17.0

Notes. SBA 7(a) Guaranty loans approved 2010Q1–2019Q4 by depository banks. Approval amount reported in 2019 dollars. The approval amount is the gross approval; the guarantee share is the SBA-guaranteed amount over the gross approval; the rate is the initial contract rate; maturity is the stated term.

A.2 Call Reports and FR Y-9C

We obtain quarterly balance-sheet and income-statement data from the FFIEC 031 Call Reports and the Federal Reserve's Consolidated Financial Statements for Holding Companies (FR Y-9C). Call Reports provide detailed financial information for individual reporting banks, including their assets, liabilities, capital, interest income and interest expense. The FR Y-9C provides corresponding information on a consolidated basis for reporting holding companies. These sources underpin our measures of banks' interest income and expense sensitivities and supply the balance-sheet characteristics used in the loan-pricing analysis.

We consolidate the data at the holding company level using institutional relationship information from the Federal Financial Institutions Examination Council (FFIEC). This unit of observation captures the scope for funding and asset allocation to be coordinated across

affiliated institutions. A holding company can raise funds on behalf of its subsidiaries, while internal capital markets allow resources to be reallocated within the group, influencing subsidiaries' lending decisions (Campello, 2002). Analysing subsidiaries in isolation could therefore omit relevant group-level funding costs and obscure the financial resources supporting their lending. Throughout the paper, "bank" refers to the consolidated entity.

We measure interest income and expense as annualised quarterly flows divided by quarterly average assets. Total interest expense includes payments on deposits, wholesale funding and other liabilities. The resulting expense sensitivity therefore captures the response of overall funding costs per dollar of assets. The regulatory reports also provide the bank size, capital and repricing information used as controls in the empirical analysis.

Following Drechsler et al. (2021), we estimate bank-specific interest expense sensitivities using a distributed-lag specification:

$$\Delta \text{IntExp}_{b,t} = \alpha_b + \eta_t + \sum_{\tau=0}^{3} \beta_{b,\tau}^{\text{exp}} \Delta \text{FedFunds}_{t-\tau} + \varepsilon_{b,t}.$$

Here, $\text{IntExp}_{b,t}$ is bank b's annualised interest expense rate: total quarterly interest expense, including payments on deposits, wholesale funding and other liabilities, divided by quarterly average assets and multiplied by four. FedFunds_t denotes the effective federal funds rate, and Δ denotes a quarterly change. The terms α_b and η_t are bank and quarter fixed effects, respectively, and $\varepsilon_{b,t}$ is an error term. The bank-specific coefficients allow funding costs to respond differently across banks, while the contemporaneous rate change and three lags capture adjustment over four quarters.

We summarise each bank's interest expense sensitivity by the sum of these coefficients,

$$\beta_b^{\text{exp}} = \sum_{\tau=0}^{3} \beta_{b,\tau}^{\text{exp}}.$$

This measure captures the sensitivity of total funding expense per dollar of assets, rather than the passthrough to deposit rates alone. We construct interest income sensitivity, β_b^{inc}, analogously, replacing the dependent variable with the quarterly change in the annualised interest income rate, defined as four times total quarterly interest income divided by quarterly average assets.

A.3 Summary of Deposits (SOD)

The FDIC Summary of Deposits (SOD) reports annual branch-level information of FDIC-insured institutions as of June 30. It records branch deposit balance, location and institutional identifiers, allowing us to characterise banks' geographic networks and measure local banking market concentration. We treat counties as local markets and branches belonging to the same holding company as a single competitor, consistent with Appendix A.2.

For county c and year t, let $N_{b,c,t}$ denote bank b's number of branches in that county and let $N_{c,t} = \sum_b N_{b,c,t}$ denote the total number of bank branches in county c. We measure county-level concentration using the Herfindahl–Hirschman index (HHI),

$$\text{HHI}_{c,t} = \sum_b \left(\frac{N_{b,c,t}}{N_{c,t}} \right)^2.$$

We then construct a bank-level measure by averaging county HHIs across the bank's branch network, $\mathcal{C}(b)$, weighting each county equally:

$$\text{HHI}_{b,t} = \frac{1}{|\mathcal{C}(b)|} \sum_{c \in \mathcal{C}(b)} \text{HHI}_{c,t},$$

This measure summarises the concentration of banking markets across the bank's geographic network, which we use later as a proxy for bank lending market power, once conditional on bank funding sensitivity β_b^{exp}.

We use $\text{HHI}_{b,t}$ as a proxy for lending-market competition conditional on interest expense sensitivity, β_b^{exp}. Local competition for borrowers can differ from the competitive conditions governing a bank's overall funding costs. Bank holding companies allocate resources across affiliated institutions through internal capital markets, allowing lending in one location to draw on funding raised elsewhere in the group (Campello, 2002). Interest expense sensitivity therefore reflects funding conditions across the bank's geographic network and wholesale markets, whereas competition for a particular borrower depends on locally available lending alternatives.[10] Figure 2 provides supporting evidence for our concentration measure: county-level branch-count HHI is positively and significantly associated with concentration measured using banks' shares of SBA loan dollars. SBA lending shares, however, reflect banks' pricing and credit-supply decisions together with borrower demand, making contemporaneous loan-share HHI potentially endogenous to the fixed–variable rate differential. Using one-year-lagged branch-count HHI mitigates this direct source of simultaneity by measuring established banking presence without incorporating current lending volumes. Its association with SBA lending concentration supports its relevance for the pricing analysis, while conditioning on β_b^{exp} accounts for differences in measured funding-cost sensitivity.

[10]The relevance of locality for lending is particularly strong in the SBA 7(a) market. Di and Pattison (2020) report that 71% of loans in their matched 2017 sample were extended to businesses located within ten miles of a branch of their lender.

B Empirical Appendix

B.1 County-level branch-count concentration

Figure 7 maps branch-count concentration, complementing the SBA loan-share map and the comparison between the two HHIs in Figure 2.

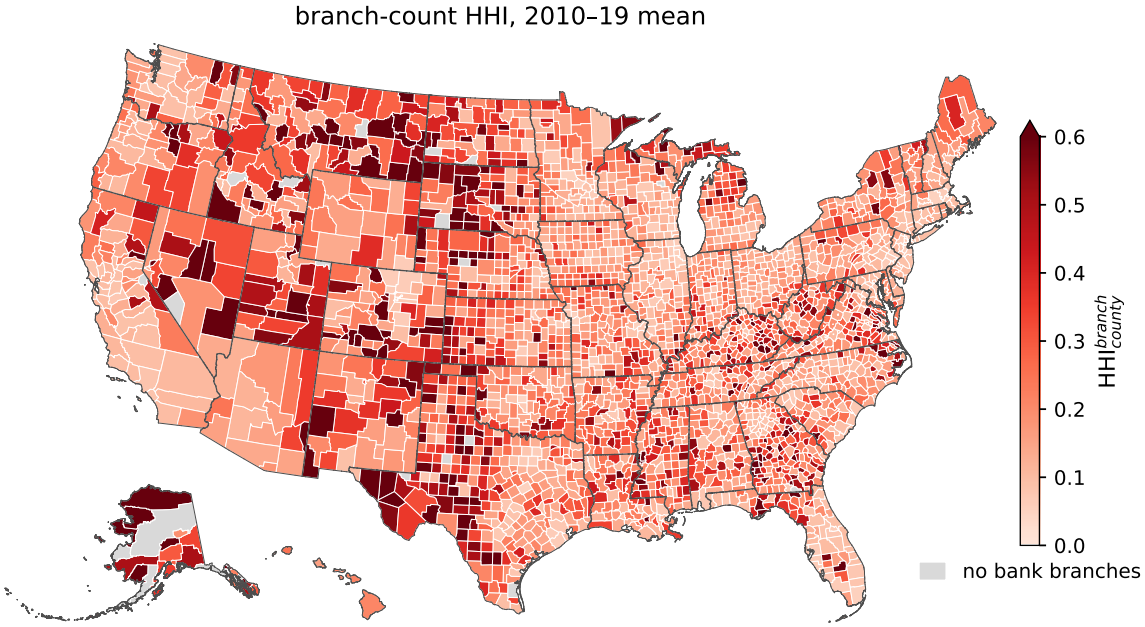


Figure 7: County-level branch-count concentration

Notes: County-level branch-count HHI is the sum of banks' squared shares of local branches. The map reports the mean annual HHI over 2010–2019. Grey counties have no bank branches. Source: FDIC Summary of Deposits.

The distinction between the two measures matters for our pricing analysis. SBA lending shares are equilibrium outcomes that reflect banks' loan pricing and supply decisions together with borrower demand. Concentration calculated from contemporaneous loan shares may therefore be jointly determined with the fixed–variable rate differential. Branch-count concentration captures the distribution of established banking presence without directly incorporating current loan volumes. We accordingly use one-year-lagged branch-count HHI to capture local market structure while reducing this direct source of simultaneity. Its positive association with SBA lending concentration supports its use as a proxy for lending-market concentration, conditional on measured funding sensitivity and the other controls. This interpretation does not require branch networks to be exogenous, but it also does not establish a causal effect of concentration on pricing.

C Proofs

C.1 Fixed-rate pricing in symmetric Cournot equilibrium

Proof of Lemma 1. Let $\mu = r_0 - \underline{r} > 0$ and $c = 1/n$. Appendix D.2 establishes the optimal deposit adjustment t_i^* in (34) after any feasible loan-portfolio profile. Hold the other banks' fixed quantities constant. Since $L_F = (1/N)\sum_j l_{F,j}$, a unilateral increase in $l_{F,i}$ lowers the market premium at rate $\partial s/\partial l_{F,i} = -\theta\sigma^2/N$. Anticipating its optimal deposit offer, bank i chooses $l_{F,i}$ to maximise

$$V_i = c\mu + l_{F,i}s(L_F, \sigma) - \mu t_i^* - \frac{\gamma\sigma^2}{2}(c - l_{F,i} - t_i^*)^2.$$

The envelope theorem gives

$$\frac{\partial V_i}{\partial l_{F,i}} = s - \frac{\theta\sigma^2 l_{F,i}}{N} + \min\{\gamma\sigma^2(c - l_{F,i}), \mu\}.$$

This derivative is continuous and strictly decreasing in the bank's own share. Away from the switching point, the second derivative is $-2\theta\sigma^2/N$ where $t_i^* > 0$ and $-\sigma^2(2\theta/N + \gamma)$ where the adjustment constraint is strictly binding at $t_i^* = 0$, establishing global strict concavity. At a symmetric allocation the first-order condition is

$$\theta\sigma^2\left[1 - (1 + 1/N)L_F\right] + \min\{\gamma\sigma^2(c - L_F), \mu\} = 0. \tag{31}$$

Its left-hand side is strictly decreasing, positive at zero and negative at one, so it has one interior root, including for real competition indices $n, N \geq 1$. The lemma's assumption that $\beta = 1 - 1/n$ at equilibrium implies $t_i^* = 0$ at this root, and the minimum in (31) equals $\gamma\sigma^2(c - L_F) = \gamma\sigma^2 G$. Substituting $s = \theta\sigma^2(1 - L_F)$ and rearranging gives (6). Strict own concavity ensures that each bank's choice is a best response to the common choice of its rivals. □

Joint characterisation of lending and deposit pass-through. To characterise the solution in both deposit-pricing regimes, define

$$L_B(n, N) = \frac{\theta + \gamma/n}{\theta(1 + 1/N) + \gamma}, \qquad L_D(N) = \frac{N}{N + 1}\left(1 + \frac{\mu}{\theta\sigma^2}\right).$$

The joint solution is

$$L_F^* = \min\{L_B, L_D\}, \qquad s^* = \theta\sigma^2(1 - L_F^*),$$
$$\beta^* = 1 - \frac{1}{n} + \left[\frac{1}{n} - L_F^* - \frac{\mu}{\gamma\sigma^2}\right]_+.$$

Writing $G_B = 1/n - L_B$, voluntary additional pass-through is inactive exactly when $\mu \geq \gamma\sigma^2 G_B$. Then equations (6) and (7) apply. Condition (35) in Appendix D.3 ensures this regime uniformly over real $n \geq 1$. For $n \geq 2$ it is automatic: the left-hand side of (31) is positive at $L_F = c$, so the root has $L_F > c$ and hence $t_i^* = 0$.

C.2 Expected utility and symmetric Cournot lending

Throughout this comparison the deposit contracts are given. For either utility function in Section 3.4, the marginal borrower's premium solves

$$u_\varrho(c-s) = \tfrac{1}{2}[u_\varrho(c+\sigma) + u_\varrho(c-\sigma)], \qquad \varrho = 2\theta(1-L).$$

Greater risk aversion reduces the certainty equivalent of the variable loan, so $s_L < 0$. Under both preferences $s(0) > 0$ and $s(1) = 0$, using the risk-neutral limit at $L = 1$.

Proof of Lemma 2. Fix rival quantities and let l be the bank's own fixed share, so $dL/dl = 1/N$. Write $\mu = r_0 - \underline{r}$. Terminal wealth in the two states is $w(\pm 1) = E + (1-\beta)\mu + ls(L) \pm \sigma(1-\beta-l)$. A marginal fixed loan pays $s + (l/N)s_L \mp \sigma$ in those states. Weighting these payoffs by marginal utility, dividing by average marginal utility, and then setting $l = L$ gives (10) with

$$H = \sigma \frac{u'(w(-1)) - u'(w(+1))}{u'(w(-1)) + u'(w(+1))}.$$

Substituting the marginal utilities and using $w(\pm 1) = W \pm \sigma G$ gives $H = \sigma \tanh(\gamma\sigma G)$ under CARA and $H = \sigma \tanh[\gamma \operatorname{arctanh}(\sigma G/W)]$ under CRRA. For CRRA we restrict attention to primitives giving positive wealth at every feasible portfolio. □

Proof of Proposition 3. Write $R(L) = s + (L/N)s_L$. For CARA, inverse demand is strictly decreasing and concave. Indeed, for $v(\varrho) = \log\cosh(\varrho\sigma)/\varrho$ and $x = \varrho\sigma$, $\varrho^3 v''(\varrho)$ equals $x^2/\cosh^2 x - 2[x\tanh x - \log\cosh x]$. It vanishes at zero and has derivative $-2x^2\tanh(x)/\cosh^2 x < 0$. The cutoff $\varrho = 2\theta(1-L)$ is affine, which gives the claimed curvature. Thus own premium revenue $ls(L)$ is strictly concave in l, and R is strictly decreasing. For CRRA these two shape restrictions are imposed. State wealth adds an affine term to own revenue, so increasing concave utility gives strict own concavity in both cases.

Now $R(0) > 0 > R(1)$, hence $L^{Cournot}$ is unique and interior. The hedging term decreases in L (under CRRA use the restriction on G/W) and has the sign of $L^{hedge} - L$. At $L^{Cournot}$ the sign of $R + H$ is the sign of $L^{hedge} - L^{Cournot}$, while at L^{hedge} it is the sign of $L^{Cournot} - L^{hedge}$. The unique root is between these shares, or equals their common value. If $L^{hedge} = 1$, the root is strictly below one. Own concavity makes this common portfolio a best response for every bank. □

Proof of Proposition 4. Let $F = s + [(1-\beta)/M]Ls_L + H$. Implicit differentiation gives $ds/d\beta = -s_L F_\beta / F_L$. Since $s_L < 0$ and $F_L < 0$, the premium falls exactly when $F_\beta = L(-s_L)/M + H_\beta > 0$. At fixed L, the CARA hedging term gives $-H_\beta = \gamma\sigma^2/\cosh^2(\gamma\sigma G)$. For CRRA put $x = \sigma G/W$. Differentiating both exposure and expected equity gives

$$-H_\beta = \frac{\gamma\sigma^2(W - \mu G)}{(W^2 - \sigma^2 G^2)\cosh^2\big(\gamma \operatorname{arctanh} x\big)}.$$

Here $W - \mu G = E + L(s+\mu) > 0$. At zero bank risk aversion $H = H_\beta = 0$, while $L^{Cournot}(-s_L)/M > 0$. Since $R'(L^{Cournot}) < 0$, an interior equilibrium and the strict inequality persist locally by continuity and the implicit function theorem. This comparison varies bank risk aversion while holding borrower preferences fixed. □

C.3 Uncertainty and the welfare cost of lending power

Proof of Proposition 5. Write $b = 1-\beta$, $h = 1/N$, and $v(a) = \log\cosh(a\sigma)/a$, with $v(0) = 0$. Premium payments cancel in the sum of monetary certainty equivalents. The risk cost at a symmetric fixed share L is therefore

$$R(L) = \frac{1}{2\theta}\int_0^{2\theta(1-L)} v(a)\,da + \frac{1}{\gamma}\log\cosh\big(\gamma\sigma(b-L)\big).$$

The first term sums the risk premia of borrowers on variable rates; the second prices bank earnings risk. Inverse demand is $s = v(2\theta(1-L))$. Write $B(L) = -R'(L) = s+\sigma\tanh(\gamma\sigma(b-L))$. Since $s_L < 0$, B is strictly decreasing. It changes sign between zero and one. The efficient share solves $B(L^\star) = 0$, while the Cournot share solves $B(L_F) + hL_F s_L(L_F) = 0$. Hence $B(L_F) > 0$, $L_F < L^\star$, and

$$0 < \Delta(\sigma) = \int_{L_F}^{L^\star} B(u)\,du \le (L^\star - L_F)hL_F[-s_L(L_F)].$$

For small σ, Taylor expansion of $\log\cosh x$ gives the mean–variance risk cost in (12) with opposite sign, up to $O(\sigma^4)$ uniformly in L. The two first-order conditions divided by σ^2 converge to their mean–variance counterparts, each with a unique root. Thus $\Delta(\sigma)/\sigma^2$ converges to the strictly positive mean–variance loss per unit variance.

For large σ, fix any $0 < \varepsilon < 1-b$. On $[b, b+\varepsilon]$ the marginal borrower's coefficient $a = 2\theta(1-L)$ stays bounded away from zero. The identity $\log\cosh x = x - \log 2 + \log(1+e^{-2x})$ gives $s = \sigma - (\log 2)/a + o(1)$ and $s_L = -2\theta(\log 2)/a^2 + o(1)$ uniformly there. At $L = b$ the bank hedging term is zero, so both marginal conditions are positive for large σ. At $L = b+\varepsilon$ the hedging term is $-\sigma + o(1)$, making B negative, and the Cournot term is negative as well. Both unique roots therefore converge to b. Since s_L remains bounded near b, the upper bound above implies $\Delta(\sigma) \to 0$. The two roots vary continuously with $\sigma > 0$ by strict monotonicity of their marginal conditions, so the loss is continuous. Its positivity and zero limits at both ends imply an interior global maximum. The argument requires neither a unique peak nor feasibility of the deposit-pricing continuation for arbitrarily large shocks. □

D Model Appendix

D.1 Ownership and the bank objective

Bank owners care about the level and variability of their terminal consumption. We assume that a bank's ownership base is proportional to its assets and that owners share earnings equally. Each owner then bears the earnings risk associated with one unit of lending. This gives the bank objective over returns on assets in Section 3.1 and, when entry is endogenous, the participation value in Section 3.6.

Ownership and preferences. Each bank i has a fixed pool $\overline{\mathcal{O}}_i$ of identical prospective owners, large enough to accommodate the bank sizes under consideration. The pools are disjoint, and an owner can participate only through its designated bank. A bank with assets $d_i > 0$ activates a subset $\mathcal{O}_i \subseteq \overline{\mathcal{O}}_i$ of mass d_i. Asset size and membership are fixed before portfolio choices and the interest-rate shock. Prospective owners outside $\mathcal{O}_i$ remain inactive.

Owners have deterministic outside terminal wealth b and mean–variance preferences over terminal consumption,

$$\mathcal{U}(c) = \mathbb{E}c - \frac{\gamma}{2}\operatorname{Var}(c), \qquad \gamma > 0.$$

The coefficient γ is common across owners and independent of bank size. An inactive owner consumes b, giving value $\mathcal{U}(b) = b$. Owners have no other exposure to the interest-rate shock.

Given bank counts. First hold the number of banks fixed and omit participation costs. Let $\Pi_i(z) = d_i\pi_i(z)$ denote total net interest earnings, where $\pi_i(z)$ is earnings per unit of assets. Equal sharing among the mass d_i of active owners gives each owner terminal consumption

$$c_i^0(z) = b + \frac{\Pi_i(z)}{d_i} = b + \pi_i(z).$$

Since outside wealth is deterministic, the owner's certainty-equivalent gain is

$$\mathcal{U}(c_i^0(z)) - \mathcal{U}(b) = \mathbb{E}\pi_i(z) - \frac{\gamma}{2}\operatorname{Var}(\pi_i(z)) \equiv W_i. \tag{32}$$

All owners therefore prefer the fixed-rate portfolio share that solves

$$\max_{l_{F,i}\in[0,1]} \left\{\mathbb{E}\pi_i(z) - \frac{\gamma}{2}\operatorname{Var}(\pi_i(z))\right\},$$

taking rival portfolios as given and anticipating the deposit continuation. This is the bank objective in Section 3.1. Mean–variance preferences are specified over owners' consumption, and equal sharing translates them into preferences over the bank's return on assets.

Summing the owners' gains gives

$$d_i W_i = \mathbb{E}\Pi_i(z) - \frac{\gamma}{2d_i}\operatorname{Var}(\Pi_i(z)),$$

where $\operatorname{Var}(\Pi_i(z)) = d_i^2 \operatorname{Var}(\pi_i(z))$. Dollar risk tolerance is d_i/γ, proportional to the mass of owners. This ownership interpretation is consistent with Wilson (1968), who shows that a syndicate's risk tolerance equals the sum of its members' risk tolerances under efficient sharing. In the symmetric benchmark, total assets are one and $d_i = 1/N$. Each bank then

has risk tolerance $1/(N\gamma)$, and aggregate risk tolerance across banks is $1/\gamma$. Increasing N divides both assets and active ownership among more banks.

Endogenous entry and participation costs. Now let banks choose whether to operate, with assets assigned according to the active bank counts as in Section 3.6. Ownership follows the same allocation rule, so a participating bank with assets d_i has active owner mass d_i. Participation costs $F > 0$, paid from owners' outside resources in terminal-value units. Active owners share the cost equally, giving each terminal consumption

$$c_i^F(z) = b + \frac{\Pi_i(z) - F}{d_i} = b + \pi_i(z) - \frac{F}{d_i}.$$

The allocation satisfies $\int_{\mathcal{O}_i}(c_i^F(z) - b)\,\mathrm{d}j = \Pi_i(z) - F$. Because the fee is deterministic, the owner's net gain is

$$\mathcal{U}(c_i^F(z)) - \mathcal{U}(b) = W_i - \frac{F}{d_i}.$$

Conditional on participation and bank size, every owner chooses the portfolio that maximises W_i. All active owners also agree on participation, which is weakly preferred when $W_i \geq F/d_i$.

Writing $c_i^F(j, z) = b$ for an inactive owner, the sum of gains across the bank's fixed pool of prospective owners is

$$V_i = \int_{\overline{\mathcal{O}}_i} \left[\mathcal{U}(c_i^F(j, z)) - b\right] \mathrm{d}j = d_i W_i - F.$$

In terms of total earnings, this gives

$$V_i = \mathbb{E}\Pi_i(z) - \frac{\gamma}{2d_i}\operatorname{Var}(\Pi_i(z)) - F. \tag{33}$$

A larger bank draws on more owners with the same individual risk tolerance and spreads the fixed cost over a larger ownership base. The common interest-rate shock affects all owners. For a given per-asset earnings distribution, dollar earnings variance grows with the square of bank size, while the coefficient on that variance falls in proportion to size.

A related entry formulation appears in Atkeson et al. (2015), who model banks as coalitions of risk-averse traders and compare per-trader certainty-equivalent gains with a fixed entry cost divided by coalition size. Their coalition sizes are given. Here active ownership follows the asset allocation implied by bank counts, so both size and ownership adjust with entry.

Symmetric entry. Consider a symmetric configuration with market size A, M local deposit communities, and n banks per community. Each bank has assets and active owner mass $d = A/(Mn)$. In the deposit retention continuation, $\beta = 1 - 1/n$. Evaluating (4) at the symmetric portfolio $l_{F,i} = L_F(n; M)$ gives per-asset earnings

$$\pi(z) = \frac{r_0 - \underline{r}}{n} + L_F(n; M)s(n; M) + \left[\frac{1}{n} - L_F(n; M)\right]\sigma z.$$

Since $\mathbb{E}z = 0$ and $\mathrm{Var}(z) = 1$, substitution into (32) yields

$$W(n; M) = \frac{r_0 - \underline{r}}{n} + L_F(n; M)s(n; M) - \frac{\gamma}{2}\sigma^2 \left[\frac{1}{n} - L_F(n; M)\right]^2.$$

This is the per-asset franchise value in (13). The first term is the expected funding advantage, the second is premium income from fixed-rate lending, and the final term is the cost of residual earnings risk borne by each owner.

Using $d = A/(Mn)$ in (33), the participating bank's net value is

$$V(n; A, M) = \frac{A}{Mn}W(n; M) - F = Av(n; M) - F, \qquad v(n; M) = \frac{W(n; M)}{Mn}.$$

The owners' participation condition is therefore equivalent to $Av(n; M) \geq F$. At fixed bank counts, a larger market expands both bank assets and active ownership, increasing total certainty-equivalent earnings relative to the fixed participation cost. Entry changes bank size and membership through the allocation rule. An additional entrant evaluates its value using its post-entry size and the reoptimised deposit and lending continuation, as in (36) and (37). This appendix verifies deposit pricing, establishes the value inequalities used in the entry analysis, and characterises the continuation following entry into a single community.

D.2 Deposit-pricing continuation

Let $\mu = r_0 - \underline{r} > 0$ and $c = 1/n$. Risk-neutral depositors rank offers $\underline{r} + \alpha(r(z) - \underline{r})$ by their expected return $\underline{r} + \alpha\mu$, so larger α is preferred even if $r(z) < \underline{r}$ in one state. For integer counts, fix any feasible loan-portfolio profile. We construct a continuation in which both eligible banks offer $\alpha = 1$ on contested accounts; the incumbent chooses its offer on uncontested accounts below. The resulting symmetric pricing conditions also define the continuous extension in n and N used in the main text.

Normalise funding expenses by the bank's committed loan book. As a reference, paying $\underline{r}$ on uncontested accounts and $r(z)$ on contested accounts gives exposure $G_i^0 = c - l_{F,i}$. Against rivals' contested-account offers of one, any unilateral bid profile changes funding expense relative to this reference by $t(r(z) - \underline{r})$ with $t \geq 0$. Losing a contested account replaces funding at the same wholesale rate; attracting a rival's account requires an offer above that rate. Raising an uncontested offer also increases t. The bank therefore chooses the net funding adjustment to maximise

$$-\mu t - \frac{\gamma\sigma^2}{2}(c - l_{F,i} - t)^2, \qquad t \geq 0.$$

Its unique optimum and resulting average pass-through are

$$t_i^* = \left[c - l_{F,i} - \frac{\mu}{\gamma\sigma^2}\right]_+, \qquad \beta_i = 1 - c + t_i^*. \tag{34}$$

Since $0 \leq t_i^* < c$, this optimum is attained by offering $\alpha = t_i^*/c < 1$ on uncontested accounts and one on contested accounts. All deposits are retained under the tie rule. The optimisation covers all unilateral bid deviations, including bids above one to obtain additional funding as a hedge. Thus these offers are best responses after every portfolio profile. We do not claim

uniqueness of all deposit-bidding equilibria.

For a given loan portfolio, (34) implies that voluntary additional pass-through is inactive exactly when $\mu \geq \gamma\sigma^2(c - l_{F,i})$. In that case uncontested accounts receive $\underline{r}$ and $\beta_i = 1 - 1/n$. Appendix D.3 gives a sufficient parameter restriction ensuring this regime at the symmetric lending equilibrium. The lending decision anticipating this deposit continuation, including the proof of Lemma 1, is established in Appendix C.1.

D.3 A sufficient condition for competition-driven deposit pass-through

Fix $M \geq 1$ and let $N = Mn$ as n varies over $[1, \infty)$. A sufficient condition for $\beta = 1 - 1/n$ at every symmetric equilibrium along this comparison is

$$\mu = r_0 - \underline{r} \geq \frac{\gamma\sigma^2\theta}{\theta(M+1) + \gamma M}. \tag{35}$$

To verify it, set $c = 1/n$ and $D = \theta(1 + 1/N) + \gamma$. Let $L_B = (\theta + \gamma c)/D$ be the candidate fixed share in (7), as characterised in Appendix C.1, and write $G_B = c - L_B$. Holding the other argument fixed gives

$$\frac{\partial G_B}{\partial c} = \frac{\theta(1 + 1/N)}{D} > 0, \qquad \frac{\partial G_B}{\partial N} = -\frac{\theta(\theta + \gamma c)}{N^2 D^2} < 0.$$

Along $c = 1/n$ and $N = Mn$, increasing n lowers c and raises N, so G_B is strictly decreasing. Its maximum occurs at $n = 1$, where

$$G_B(1, M) = \frac{\theta}{\theta(M+1) + \gamma M}.$$

Condition (35) therefore implies $\mu \geq \gamma\sigma^2 G_B$ for every $n \geq 1$, giving $L_F^* = L_B$ and zero additional pass-through. It also permits equality at $n = 1$; the envelope condition remains valid at that switching point. This is an equilibrium restriction, not a claim that additional pass-through is never optimal after a deviation in loan portfolios; those deviations are covered by (34).

Equivalently, the condition requires market breadth to satisfy

$$M \geq \underline{M} \equiv \frac{\theta}{\theta + \gamma}\left(\frac{\gamma\sigma^2}{\mu} - 1\right).$$

The negative premium–beta result in Proposition 2 also requires $M < \overline{M} \equiv \theta^2/[\gamma(\theta + \gamma)]$. Thus sufficient joint restrictions are

$$\max\{1, \underline{M}\} \leq M < \overline{M}.$$

For continuous market breadth this interval is nonempty exactly when $\overline{M} > 1$ and $\mu > \gamma^2\sigma^2/(\theta + \gamma)$; when M is a literal number of markets, the interval must contain an integer. A larger M helps keep voluntary deposit hedging inactive but can reverse the premium slope, so a lower bound on M alone does not establish the negative relationship. If $\mu \geq \gamma\sigma^2$, then $\underline{M} \leq 0$ and no additional lower bound beyond $M \geq 1$ is needed.

For the entry extension only, the additional assumption $\mu > \gamma\sigma^2$ makes $t_i^* = 0$ after every feasible portfolio profile, including when a local market has one bank. The same argument applies with heterogeneous counts using $c_i = 1/n_i$ and each bank's own asset normalisation. This supplies the continuation used in the entry derivations below.

D.4 Franchise value and the participation comparative statics

Maintain the entry extension's additional assumption $\mu \equiv r_0 - \underline{r} > \gamma\sigma^2$. Let $x = 1 - \beta$, and let $\widehat{W}(\beta, N)$ be symmetric per-asset value when beta and lender counts are treated as separate arguments. The portfolio condition implies

$$\widehat{W} = x\mu - \frac{\gamma}{2}\sigma^2 x^2 + \sigma^2 \left(\frac{\theta}{N} + \frac{\gamma}{2}\right) (L_F)^2.$$

The first two terms are nonnegative because $\mu > \gamma\sigma^2$ and $x \in [0, 1]$, while the last is positive. Using $L_{F,\beta} = -\gamma/D$ and $L_{F,N} = \theta L_F/(N^2 D)$ gives

$$\begin{aligned}
\widehat{W}_\beta &= -\mu + \gamma\sigma^2 x - \frac{2\gamma\sigma^2(\theta/N + \gamma/2)L_F}{D} < 0, \\
\widehat{W}_N &= -\frac{\theta^2\sigma^2(L_F)^2(1 - 1/N)}{N^2 D} \leq 0.
\end{aligned}$$

Along $\beta = 1 - 1/n$, $N = Mn$,

$$W_n = \frac{\widehat{W}_\beta}{n^2} + M\widehat{W}_N < 0, \qquad v_n = \frac{W_n}{Mn} - \frac{W}{Mn^2} < 0.$$

The per-asset value remains bounded as $n \to \infty$, so $v \to 0$. For $AW(1; M)/M > F$, continuity and monotonicity yield a unique real $n > 1$ solving (14). Log differentiation yields

$$\frac{d\log n}{d\log A} = \frac{1}{1 - nW_n/W} \in (0, 1).$$

Since $\beta = 1 - 1/n$ and $N = Mn$, this elasticity implies (15) and completes the proof of Proposition 6. Entry grows less than proportionally with market size, but more local rivals bid for deposits and more lenders compete for borrowers.

D.5 Entry with integer bank counts

A single local market: exact integer entry. For $M = 1$, all banks reoptimise deposit and lending choices when entry raises n to $n + 1$. An active integer equilibrium with $n \geq 1$ requires

$$Av(n; 1) \geq F \geq Av(n + 1; 1). \tag{36}$$

Incumbents cover F and another entrant cannot profit. If $Av(1; 1) < F$, the market is inactive. At equality either zero or one bank can operate.

Define $A_m = F/v(m; 1)$. By Appendix D.4, these thresholds are strictly increasing and unbounded. A count $m \geq 1$ satisfies (36) exactly when $A \in [A_m, A_{m+1}]$. Thus market growth raises the bank count in steps, with two adjacent counts possible at a threshold.

Both deposit beta and the number of lenders are nondecreasing in A, and Proposition 2 gives the direction of the premium change across counts.

Several local markets: unilateral entry. For $M > 1$, (14) approximates symmetric participation. An individual entrant changes only its home community's count. Banks then differ in size and beta, so valuing entry requires reoptimising their portfolios.

Let $\mathbf{n} = (n_1, \ldots, n_M)$ be a positive integer vector of local counts, with $N = \sum_{j=1}^{M} n_j$, and let $v_j(\mathbf{n})$ denote a bank's risk-adjusted continuation value in community j, per unit of aggregate market size. Appendix D.6 derives this continuation. An active integer entry equilibrium satisfies, for each community,

$$Av_j(\mathbf{n}) \geq F, \qquad Av_j(\mathbf{n} + \mathbf{e}_j) \leq F, \tag{37}$$

where $\mathbf{e}_j$ adds one bank in its home community j. Banks compare participation with staying out ex ante, anticipating portfolio adjustment. After entry the cost is sunk.

From symmetry the incumbent's value per unit of market size is $v(n; M)$. Entry in the first community gives $(n+1, n, \ldots, n)$, whereas $(n+1, \ldots, n+1)$ requires entry everywhere. Appendix D.7 illustrates both premium slopes under unilateral entry.

D.6 Continuation equilibrium with unequal local counts

Take a vector $\mathbf{n} = (n_1, \ldots, n_M)$ with every $n_j \geq 1$ and $N = \sum_{j=1}^{M} n_j$. A bank in community j has assets $d_j = A/(Mn_j)$, global loan-capacity weight $w_j = d_j/A = 1/(Mn_j)$, and deposit beta $\beta_j = 1 - 1/n_j$. Banks in that community have identical primitives. Let $l_{F,j}$ denote their common fixed share. Aggregate fixed lending as a fraction of total credit is $L_F = M^{-1} \sum_j l_{F,j}$.

Write $h = s/\sigma^2$. The portfolio condition, including boundary choices, gives

$$l_{F,j}(h) = \min\left\{1, \max\left\{0, \frac{h + \gamma/n_j}{\gamma + \theta/(Mn_j)}\right\}\right\}. \tag{38}$$

Market clearing requires

$$h = \theta\left[1 - \frac{1}{M}\sum_{j=1}^{M} l_{F,j}(h)\right].$$

The left side minus the right side is continuous and strictly increasing. It is negative at $h = 0$, since $l_{F,j}(0) < 1$ for every j, and positive at $h = \theta$. Thus there is a unique root in $(0, \theta)$. Equation (38) gives global best responses by own concavity.

Uniqueness also holds without imposing within-community symmetry. Index all banks by i, with capacity weights $w_i > 0$ summing to one and $x_i = 1 - \beta_i$. The lending game has the strictly concave weighted potential

$$\mathcal{P}(\mathbf{l}) = \theta \sum_i w_i l_i - \frac{\theta}{2}\left(\sum_i w_i l_i\right)^2 - \frac{\theta}{2}\sum_i w_i^2 l_i^2 + \gamma \sum_i w_i x_i l_i - \frac{\gamma}{2}\sum_i w_i l_i^2.$$

Its derivative divided by w_i equals the derivative of bank i's objective divided by σ^2. The Hessian is $-\theta \mathbf{w}\mathbf{w}' - \mathrm{diag}(\theta w_i^2 + \gamma w_i)$, which is negative definite. Hence the Nash portfolio allocation is unique and symmetric among banks with identical local primitives.

With the solution $h(\mathbf{n})$ and shares $l_{F,j}(\mathbf{n})$, define

$$v_j(\mathbf{n}) = \frac{1}{Mn_j}\left\{\frac{r_0 - \underline{r}}{n_j} + \sigma^2 l_{F,j} h - \frac{\gamma}{2}\sigma^2\left(\frac{1}{n_j} - l_{F,j}\right)^2\right\}. \tag{39}$$

A bank's total continuation value before the fee is Av_j. Recompute (38)–(39) at $\mathbf{n} + \mathbf{e}_j$ to evaluate entry. The positive-count entry conditions (37) follow. For symmetric configurations, all communities face the same entry test. We analyse active configurations rather than characterise all equilibria with inactive communities.

D.7 Numerical illustration of integer entry

Table 10 reports symmetric configurations satisfying participation and no-profitable-entry conditions, using a constant participation cost F. The entrant's continuation is evaluated at $(n+1, n, \ldots, n)$, with all bank portfolios reoptimised. The examples establish that both premium-slope regimes can occur with unilateral entry.

Breadth M	Size A	Local banks n	Lenders N	Deposit beta	Premium (bp)
1	2	12	12	0.9167	59.15
1	4	18	18	0.9444	51.92
12	2	2	24	0.5000	32.12
12	4	3	36	0.6667	33.50

Table 10: Market size and integer entry. Parameters are $\theta = 1600$, $\gamma = 160$, $\sigma = 0.005$, $r_0 = 0.04$, $\underline{r} = 0.01$, and $F = 0.001$. Each row satisfies participation and no-profitable-entry conditions after recomputing all loan portfolios following entry into one community. These are illustrative equilibria, not a calibration.

For all four rows in Table 10, evaluating these continuation values gives strictly positive incumbent net values and strictly negative entrant net values. These examples illustrate the entry conditions; they do not establish a general existence or uniqueness theorem for the multi-community entry game.

E Investment Model: Derivations and Checks

The quantitative exercises use one joint equilibrium with exact CARA financial preferences, endogenous project sizes and household saving. This appendix records their demand, pricing and market-clearing conditions, numerical checks, and welfare and conditional-efficiency calculations.

E.1 Investment and pricing under CARA

For the derivations, abbreviate hours per unit invested as $\ell(w_1) = [(1-\alpha)\bar{p}A_1/w_1]^{1/\alpha}$, so optimal hours are $h = \ell(w_1)k$. Expected surplus R is defined in (18). For borrower risk aversion t, optimised certainty equivalents are

$$U_F(s) = \max_{k\geq 0}\{(R - s + v_F)k - \kappa k^2/2\}, \qquad U_V(t) = \max_{k\geq 0}\{Rk - \kappa k^2/2 - t^{-1}\log\cosh(t\sigma k)\}.$$

The fixed-contract benefit is nonpecuniary and measured in units of profit. It affects contract and scale choices but is not a cash receipt in the goods accounts. Differentiating the variable-loan objective with respect to k gives $R - \kappa k - \sigma\tanh(t\sigma k)$. Its second derivative is $-\kappa - t\sigma^2\operatorname{sech}^2(t\sigma k) < 0$, so the first-order condition uniquely determines an interior optimum. Together with the fixed-loan condition,

$$k_F = \frac{R - s + v_F}{\kappa}, \qquad \kappa k_V(t) + \sigma\tanh(t\sigma k_V(t)) = R.$$

The optimised floating value decreases with t, so $U_V(t^*) = U_F(s)$ defines the cutoff separating floating from fixed borrowers.

At the indifferent type $t^* = \theta^{m,*} > 0$, $U_F(s) = \kappa k_F^2/2$. Substituting the variable-loan first-order condition into $U_V(t^*)$ and imposing indifference gives, with $x = t^*\sigma k_V(t^*) > 0$,

$$\frac{\kappa}{2}\left[k_F^2 - k_V(t^*)^2\right] = \frac{x\tanh x - \log\cosh x}{t^*} > 0.$$

The numerator is zero at $x = 0$ and has derivative $x\operatorname{sech}^2 x > 0$. Hence $k_F > k_V(\theta^{m,*})$ on the interior branch with positive rate uncertainty.

Write $z = (\log t^* - \mu^m)/\tau^m$, $f = 1 - \Phi(z)$ and $\mathcal{C}(t,k) = t^{-1}\log\cosh(t\sigma k)$. Along the cutoff curve,

$$k_F(z) = \sqrt{2U_V(t^*)/\kappa}, \qquad s(z) = R + v_F - \kappa k_F(z).$$

The envelope theorem yields

$$s_z = \frac{\tau^m}{k_F}\{\sigma k_V(t^*)\tanh(t^*\sigma k_V(t^*)) - \mathcal{C}(t^*, k_V(t^*))\} > 0.$$

Per entrepreneur, fixed and floating quantities and their cutoff derivatives are

$$q(z) = fk_F, \qquad q_z = -\phi(z)k_F - fs_z/\kappa,$$
$$v(z) = \int_{-\infty}^{z} k_V(e^{\mu^m + \tau^m u})\phi(u)\,du, \qquad v_z = k_V(t^*)\phi(z).$$

Thus $Q_F^m = \varphi^m q$, $Q_V^m = \varphi^m v$, $s_Q^m = s_z/(\varphi^m q_z)$ and $\frac{\partial Q_V^m}{\partial Q_F^m} = v_z/q_z$. At the indifferent type

$k_V(t^*) < k_F$, so $-1 < \frac{\partial Q_V^m}{\partial Q_F^m} < 0$: some reduction in fixed credit is a reduction in project scale, rather than a one-for-one switch to floating finance.

Bank certainty equivalents and pricing. For a symmetric bank let $d^m = \mathcal{D}^m/n^m$, $G^m = 1 - \beta^m - Q_F^m/\mathcal{D}^m - \eta^m$ and

$$\bar{\pi}^m = (1-\beta^m)(r_0 - \underline{r}) + \frac{Q_F^m s^m + b_V Q_V^m}{\mathcal{D}^m} - a\eta^m.$$

Operating earnings per deposit are $\bar{\pi}^m + \sigma G^m z$. CARA over return on equity with coefficient γ and $E^m = ed^m$ therefore gives

$$W^m = \bar{\pi}^m - \frac{1}{\widetilde{\gamma}} \log\cosh(\widetilde{\gamma}\sigma G^m), \qquad \widetilde{\gamma} = \gamma/e.$$

Maximising W^m over $\eta^m \geq 0$ gives

$$W_\eta^m = -a + \sigma \tanh(\widetilde{\gamma}\sigma G^m), \qquad W_{\eta\eta}^m = -\widetilde{\gamma}\sigma^2 \operatorname{sech}^2(\widetilde{\gamma}\sigma G^m) < 0.$$

Thus the hedge is unique conditional on lending. For $a < \sigma$,

$$\eta^m = \max\left\{1 - \beta^m - \frac{Q_F^m}{\mathcal{D}^m} - \frac{\operatorname{atanh}(a/\sigma)}{\widetilde{\gamma}\sigma},\, 0\right\};$$

for $a \geq \sigma$, it is zero. The lending objective need not be globally concave after borrower investment and hedge choice are incorporated.

Under the restricted-access construction below, a unilateral change in fixed lending has price derivative Js_Q^m at symmetry, while the maintained variable-loan allocation gives $\frac{\partial q_V^m}{\partial q_F^m} = \frac{\partial Q_V^m}{\partial Q_F^m}$. Differentiating the bank's certainty equivalent, applying the envelope theorem to its hedge choice, yields

$$0 = s^m + J\frac{Q_F^m}{n^m} s_Q^m + b_V \frac{\partial Q_V^m}{\partial Q_F^m} + \sigma \tanh(\widetilde{\gamma}\sigma G^m).$$

Rearranging gives the pricing condition in Section 4.2.

Restricted lender access and the continuous approximation. For integer $J \geq 1$ and bank counts divisible by J, divide market m into J identical lending segments indexed by ℓ. Each has borrower mass φ^m/J, the same borrower type distribution, and $n^m/J \geq 1$ banks. Each bank serves one segment; borrowers cannot borrow outside it. Deposit search remains market-wide, so $\beta^m = p(n^m; \zeta^m)$ and deposits per bank remain $d^m = \mathcal{D}^m/n^m$. Conditional on loan terms, borrower preferences, technology and investment choices are unchanged.

Let $Q_j^m(s; R)$ denote demand from the full borrower population under contract $j \in \{F, V\}$. Segment demand is $Q_{j,\ell}^m(s; R) = Q_j^m(s; R)/J$, so its fixed-loan inverse demand is $s_\ell^m(Q; R) = s^m(JQ; R)$. At symmetry,

$$Q_{F,\ell}^m = Q_F^m/J, \qquad q_F^m = \frac{Q_F^m/J}{n^m/J} = \frac{Q_F^m}{n^m}, \qquad \frac{\partial s_\ell^m}{\partial q_F^m} = Js_Q^m.$$

The last equality holds with segment rivals' quantities fixed. Thus ordinary Cournot competition generates the markup term $J(Q_F^m/n^m)s_Q^m$. For variable lending, the common population scaling cancels:

$$\frac{\partial Q_{V,\ell}^m}{\partial Q_{F,\ell}^m} = \frac{\partial Q_V^m/\partial s}{\partial Q_F^m/\partial s} = \frac{\partial Q_V^m}{\partial Q_F^m}.$$

We retain the allocation rule that the bank changing fixed lending absorbs this marginal change in segment variable lending; segmentation alone does not derive that rule. Per-bank lending, deposits and equity are unchanged at symmetry, preserving the service-income and hedging terms. This reproduces the pricing condition above.

The calibration extends these symmetric equations continuously to non-integer J, just as it uses continuous bank counts. Non-integer J measures effective segmentation, with $n^m/J \geq 1$ on the reported equilibrium paths; it is not an exact finite-bank partition. Holding segmentation fixed as entry adjusts preserves the existing symmetric demand, profit, entry and resource equations, and hence the calibration and reported counterfactual solutions. Entry is symmetric across identical segments with the same market-specific hurdle F^m. Market-wide HHI remains $1/n^m$; within-segment HHI is J/n^m, so the calibration maps observed HHI to the broader market. This derivation establishes the symmetric first-order conditions, not global optimality of bank choices or stability against asymmetric segment entry.

E.2 Household saving and implementation

Household utility is log in current consumption and linear in terminal consumption, as in (22). Deposit relationship weights δ^m are independent of borrower populations. Expected gross deposit return is

$$\mathbb{E}R_d = 1 + \sum_m \delta^m\{(1-\beta^m)\underline{r} + \beta^m r_0\}.$$

The household conditions are

$$C_0^{-1} = \xi\mathbb{E}R_d, \qquad \chi_0 L_0^\nu = w_0/C_0, \qquad \chi_1 L_1^\nu = w_1.$$

These conditions are imposed in both the monetary and capacity experiments. The terminal wage is determined before the rate fixing but is not held at its baseline across counterfactuals.

The benchmark is conditional on an exogenous real interbank rate. A government facility can implement it by intermediating matching surplus and deficit bank positions at the same state-contingent floating payoff. Since bank balance sheets and current goods-market clearing imply $\mathcal{D} = I$, the facility's net position and statewise tax requirement are zero. This implementation relies on inherited current retail supply contracts and is not a claim that an arbitrary real rate clears a fully flexible-price economy.

Retail prices and baseline normalisations. The final-good aggregator has elasticity ε across retail varieties. Flexible terminal pricing implies intermediate-good price $\bar{p} = (\varepsilon-1)/\varepsilon$, retail output $Y_1 = A_1\ell^{1-\alpha}I$, and household-owned retail dividends Y_1/ε. Current retail supply prices are set before the surprise. At the initial allocation, the optimal markup gives $p_0^b = \bar{p}$; after the surprise, $p_0 = w_0/A_0$ adjusts while the inherited retail contract remains in force.

We normalise $I^b = L_0^b = L_1^b = 1$, implying $\ell^b = 1$. Given fitted baseline project surplus R^b, the productivity and preference scales are

$$A_1 = \frac{1 + r_0^b + R^b}{\alpha \bar{p}}, \quad \chi_1 = \frac{1-\alpha}{\alpha}(1 + r_0^b + R^b), \quad A_0 = C_0^b + 1, \quad \chi_0 = \bar{p} A_0 / C_0^b.$$

These date-specific scales support the normalised allocation while accommodating the different technologies and marginal utilities of consumption at the two dates. They are fixed in all counterfactuals, so subsequent deposits, investment, wages and employment remain equilibrium outcomes.

Deposit search and counts. The deposit-search rule is $\beta^m = \zeta^m(n^m - 1)/[1 + \zeta^m(n^m - 1)]$. Consequently observed baseline betas and effective counts $n^{m,b} = 1/H^{m,b}$ imply $\zeta^m = \beta^{m,b}/[(1 - \beta^{m,b})(n^{m,b} - 1)]$. Counts are continuous competition indices. At baseline, the calibration's positive gap $r_0 - \underline{r} > \sigma$ makes deposits dominate the outside return state by state when $\beta^m > 0$.

E.3 Equilibrium and resource accounts

Terminal labour supply and labour demand imply

$$\ell = I^{-\nu/(\nu+\alpha)}, \qquad 1 + r_0 + R = (1 + r_0^b + R^b) I^{-p}, \qquad p = \frac{\nu(1-\alpha)}{\nu + \alpha}.$$

Conditional on counts, the real equilibrium can therefore be expressed using fixed and variable credit per entrepreneur, $q^m = Q_F^m/\varphi^m$ and $v^m = Q_V^m/\varphi^m$, as

$$\log \frac{1 + r_0 + R}{1 + r_0^b + R^b} + p \log \left[\sum_m \varphi^m \{q^m + v^m\} \right] = 0. \tag{40}$$

This condition and the two pricing equations determine the project surplus and borrower cutoffs. Free entry adds two equations $\mathcal{D}^m W^m / n^m = F^m$ for the monetary experiment. The capacity experiment instead fixes counts at their baseline and replaces $\widetilde{\gamma}$ by $\widetilde{\gamma}/c$. Household deposits always satisfy $\mathcal{D} = I$.

Let $J_2 = \sum_m \varphi^m \mathbb{E}(k^m)^2$, $Q_V = \sum_m Q_V^m$ and $K_B = \sum_m n^m F^m$. Aggregate bank-owner net payouts after overhead, entrepreneur income and household consumption are

$$\begin{aligned}
C_1^B(z) &= \sum_m \mathcal{D}^m \{\bar{\pi}^m + \sigma G^m z\} - K_B, \\
C_1^E(z) &= RI - \frac{\kappa}{2} J_2 - \sum_m s^m Q_F^m - \sigma z Q_V, \\
C_1(z) &= w_1 \ell I + R_d(z) I + Y_1/\varepsilon.
\end{aligned}$$

The common bank benefit is accounted for as net ancillary-service output $b_V Q_V$ accruing to bank owners. The borrower convenience benefit is nonpecuniary. Summing the cash budgets yields, in each rate state,

$$C_1(z) + C_1^E(z) + C_1^B(z) + \frac{\kappa}{2} J_2 + K_B + \mathcal{H} + \Lambda(z) = Y_1 + b_V Q_V.$$

The hedge resource cost is $\mathcal{H} = a\sum_m \mathcal{D}^m\eta^m$ and $\Lambda(z) = \sigma z\sum_m \mathcal{D}^m\eta^m$ is the net payment to outside counterparties. Its expectation is zero, and counterparties are not included in household welfare. Financial-owner payouts are net income: fixed outside wealth can cover adverse-state losses and cancels from the resource identity. Current resources satisfy $C_0+I = A_0L_0$. Local interbank positions $\mathcal{D}^m - I^m$ sum to zero. Thus changes in deposits are endogenous equilibrium changes in household saving, rather than an exogenous funding restriction.

E.4 Numerical checks

The calibration matches the eight point targets to numerical precision and chooses an HHI decline of 0.065233 inside the accepted interval. Equal borrower means restrict $\mu^H = \mu^L + ((\tau^L)^2 - (\tau^H)^2)/2$. Borrower integration is checked at 480 nodes against the calibration quadrature, and independent equilibrium solves reproduce the calibrated allocations. Goods accounts include both hedge costs and state-contingent external payments.

For reported equilibria we numerically search each bank's feasible fixed-lending deviations, optimising its outside hedge at every choice and enforcing nonnegative variable lending. This checks global bank choice on the maintained symmetric-rivals problem, rather than relying only on first-order conditions. It is numerical evidence, not an analytic global-optimality or uniqueness theorem.

The deposit-pricing incentive check is $r_0 - \underline{r} \geq \sigma\tanh(\widetilde{\gamma}\sigma G^m)$. The right-hand side equals a with positive outside hedging and is at most a otherwise. The fitted $a = 9.48$ bp is below the 59 bp funding gap. This checks the reported portfolio, not deposit continuation following every deviation.

Outside-return sensitivity. Holding the calibration fixed, let $\Delta\underline{r} = \lambda_{\text{out}}\Delta r_0$ for the 50 bp benchmark increase. For $\lambda_{\text{out}} = 0, 0.5, 1$, aggregate premium changes are -3.00, -2.21 and $+0.06$ bp, fixed-share changes are $+1.30$, $+0.67$ and -0.03 pp, investment changes are -0.673, -0.677 and -0.687%, and HHI changes are -0.0652, -0.0401 and $+0.0012$. Allowing the outside return to rise reduces the funding-margin gain and entry; with equal rate increases, bank counts fall slightly. These are conditional rate comparisons, not an estimated policy rule. All pass numerical bank-choice checks; none has a high-beta outside hedge. The low-beta hedge is zero for $\lambda_{\text{out}} = 0, 0.5$ and positive for $\lambda_{\text{out}} = 1$.

E.5 Conditional efficiency and access to lenders

For each equilibrium, hold type-specific project sizes $k^m(t)$, wages, deposits, counts and deposit contracts fixed. Let $\iota^m(t) \in \{0, 1\}$ indicate fixed finance, with

$$Q^m = \varphi^m \int \iota^m(t)k^m(t)\,\mathrm{d}F_\theta^m(t).$$

Optimise contract assignment and outside hedge ratios $\eta^m \geq 0$ to maximise combined borrower and bank monetary CE with equal dollar weights. Dropping allocation-invariant terms

gives

$$\mathcal{S}(\iota,\eta)=\sum_m \varphi^m \int \left[\{b_V+(v_F-b_V)\iota^m(t)\}k^m(t)-\frac{1-\iota^m(t)}{t}\log\cosh(t\sigma k^m(t))\right]\mathrm{d}F_\theta^m(t)$$
$$-\sum_m \mathcal{D}^m\left\{a\eta^m+\frac{1}{\widetilde{\gamma}}\log\cosh\left[\widetilde{\gamma}\sigma\left(1-\beta^m-\frac{Q^m}{\mathcal{D}^m}-\eta^m\right)\right]\right\}.$$

Premiums cancel as transfers. The objective includes service output, nonpecuniary contract benefits, borrower and bank risk costs, and real hedge costs. It excludes changes in production, project sizes, entry, household allocations and outside counterparties' welfare.

For fixed project sizes, the direct gain per unit from fixing is

$$v_F-b_V+\frac{\log\cosh(t\sigma k^m(t))}{tk^m(t)}.$$

Its monotonicity in exposure $tk^m(t)$ permits a threshold assignment, allowing for the project-size discontinuity at the original cutoff. We optimise each threshold and use the hedge formula in Appendix E.1. The conditional gap is the resulting CE gain per unit of equilibrium investment. Outside hedges are zero at the computed efficient assignments: fixed lending provides the preferred hedge once the pricing distortion is removed.

Table 11: Conditional risk-sharing efficiency

σ (bp)	Market fixed share (%)	Efficient fixed share (%)	Gap (bp)
50	18.59	38.32	1.349
75	17.93	53.73	3.421

Notes: The benchmark conditions on each equilibrium allocation. The larger gap at 75 bp is a comparison of these allocations, not a global monotonicity result.

Separate lender-access comparison. At $\sigma = 75$ bp, reduce J by 25%, from 3.94 to 2.96, holding other primitives and entry hurdles fixed but allowing entry, wages, saving and investment to adjust. Relative to the original-J selected equilibrium, the premium falls by 10.79 bp, the fixed share rises by 20.68 pp and investment rises by 0.05%. The conditional gap falls from 3.42 to 1.50 bp, while average entrepreneur CE falls by 0.17% and household welfare rises by 0.05%. Premiums use the reference equilibrium's loan weights. This checked counterfactual has no high-beta outside hedge and is separate from the both-market-hedging selection used for the uncertainty comparison; it does not establish a transition induced by an access reform. Policy implementation costs are excluded, and the welfare measures are not aggregated.